METHODS, DRAWN.

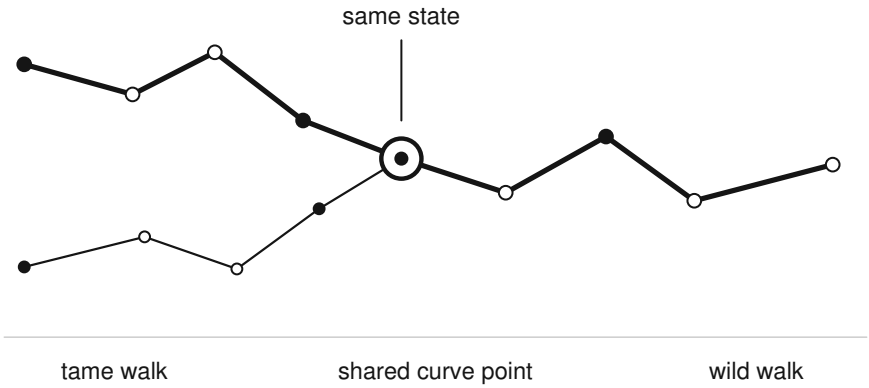
225 monochrome mechanism studies

A visual companion to the Bitcoin research inventory

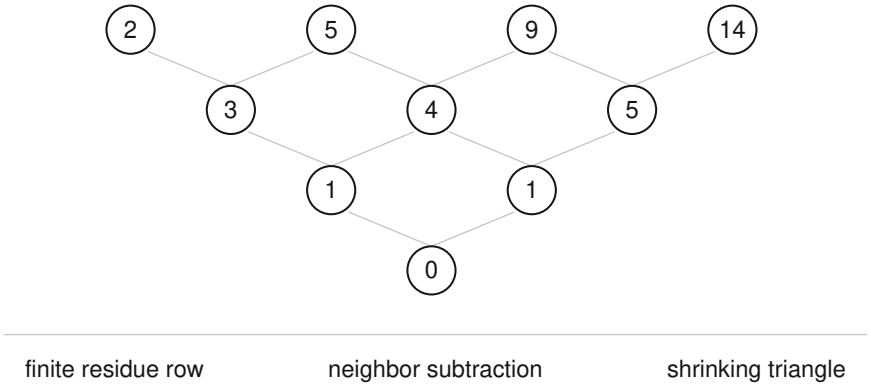
Black marks show the active object or route. Gray marks show context, alternatives, or supporting structure. Thin rules show relationships; heavier strokes show the operation being emphasized.

Each method retains its original ID, evidence label, and report finding. Schematic positions and weights are illustrative. The drawings describe mechanisms, including hypotheses and unsuccessful routes; the source finding states what was actually established.

M022 / Tame/wild collision recovery



M202 / Finite modular difference triangles



Visual vocabulary

active operation

context or relationship

states, values, or candidate positions

Vector drawings • 14 method families • 225 source-linked entries

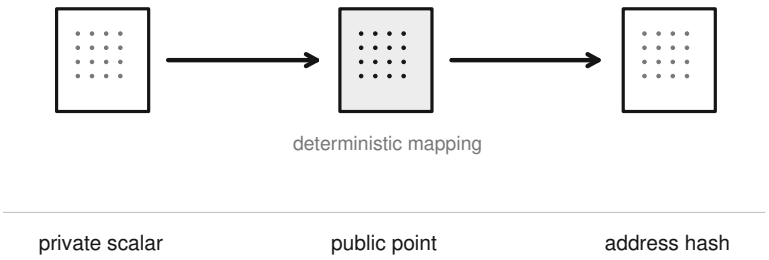
CONTENTS

Methods are grouped exactly as in the source inventory.

01	Verification, sampling and live search	20 methods	**3**
02	Kangaroo, interval ECDLP and execution backends	17 methods	**8**
03	CTheory decompositions and scalar geometry	18 methods	**13**
04	Predictors, held-out validation and corrective audits	16 methods	**18**
05	Root-difference, carry, interpolation and near-miss routes	15 methods	**22**
06	Generator reconstruction, masking and deterministic sequences	16 methods	**26**
07	Sparse keys, bit walks and synthesis orbits	13 methods	**30**
08	HP-RTE, topology and pattern-avoidance experiments	15 methods	**34**
09	ECDSA, nonce relations and signature-data searches	15 methods	**38**
10	On-chain, metadata and clue relationships	15 methods	**42**
11	Weak-entropy wallet and PRNG research	22 methods	**46**
12	Exploratory number theory, statistics and literature-inspired probes	19 methods	**52**
13	Modular differences and general transformation frameworks	11 methods	**57**
14	Shor, classical factoring and quantum-resource research	13 methods	**60**

M001 / SCHEMATIC Executed record

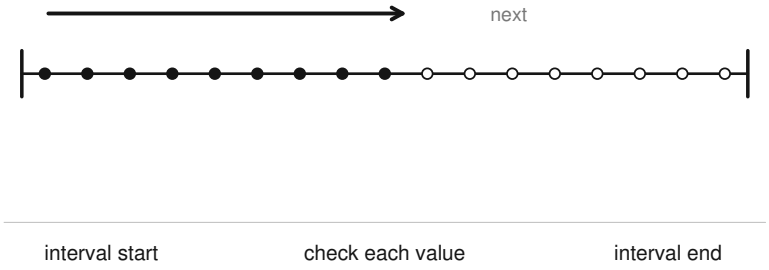
Public puzzle corpus verification



Finding: Recorded public scalars are checked against secp256k1 points and HASH160; validates the dataset, not a prediction.

M002 / SCHEMATIC Executed record

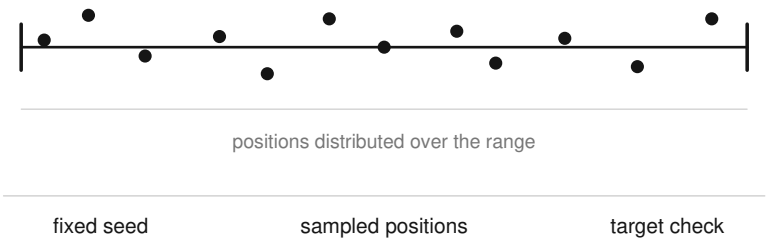
Small-range exhaustive address search



Finding: Puzzles 1-10 scanned completely; each yielded the expected published key. Small-range recovery is not an unknown-key breakthrough.

M003 / SCHEMATIC Executed record

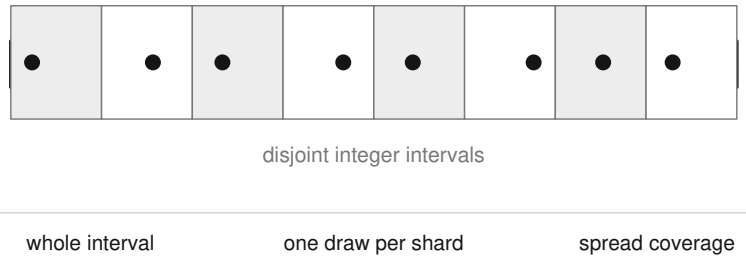
Full-range seeded sampling backtests



Finding: Actual address checkers recovered small published puzzles; seeded search order does not establish predictive narrowing.

M004 / SCHEMATIC Executed record

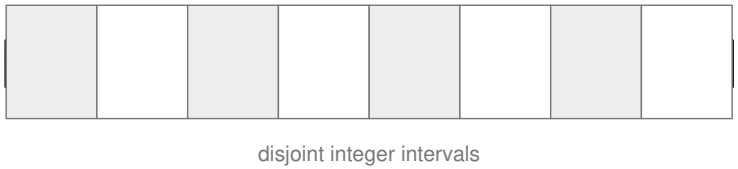
Uniform within-shard representatives



Finding: Samples one integer per disjoint interval; representative checks do not exhaust their shard.

M005 / SCHEMATIC Executed record

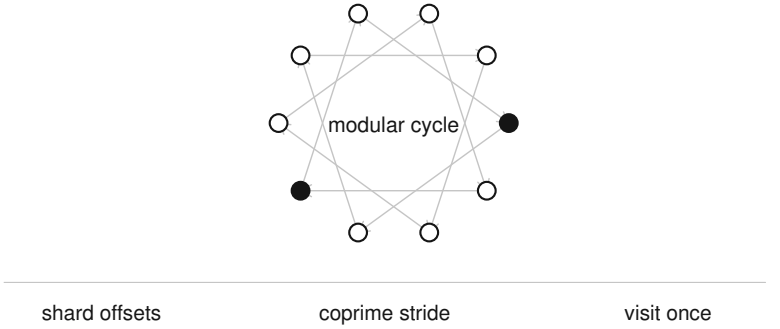
Exact integer partition coverage



Finding: Gap-free inclusive intervals replace biased single-key records. Exhaustive small-range checks verify the partition arithmetic.

M006 / SCHEMATIC Implemented

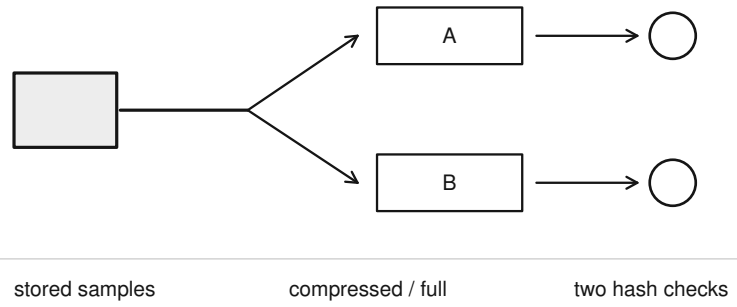
No-repeat affine shard permutations



Finding: Coprime strides enumerate each shard without a visited-key set; retains full support.

M007 / SCHEMATIC Executed record

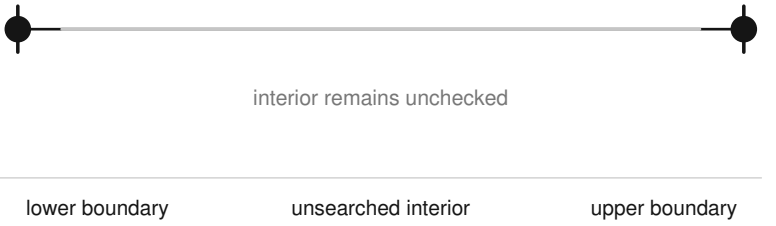
One-million sample-key verification



Finding: One million explicit samples checked in both public-key serializations; no target match.

M008 / SCHEMATIC Executed record

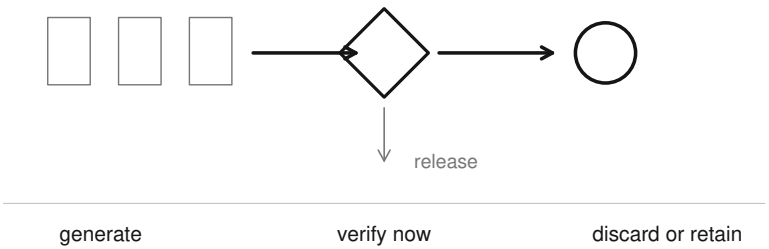
Range-start and range-end verification



Finding: Two million endpoint entries checked in both address formats; no target match. Interiors were not searched.

M009 / SCHEMATIC Executed record

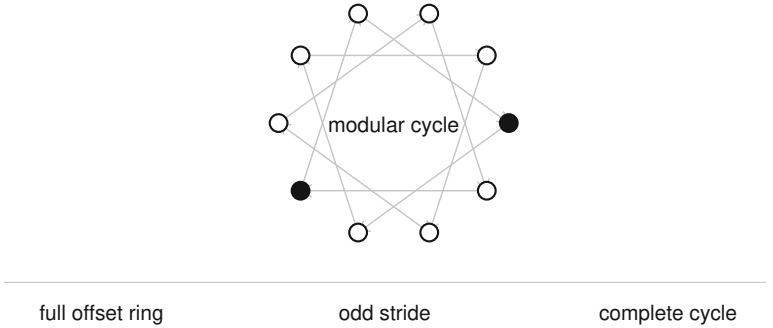
Generate-and-verify streaming batches



Finding: One million additional samples checked immediately without a large dataset; no matches in the saved batch.

M010 / SCHEMATIC Live snapshot

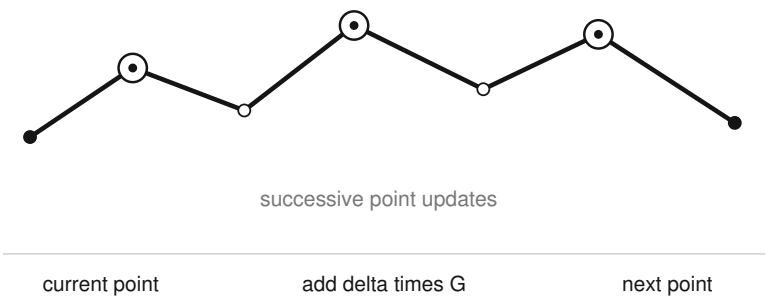
Continuous affine full-support search



Finding: Checkpointed local worker checks an affine permutation of the 70-unknown-bit interval. Status is mutable and must be time-stamped.

M011 / SCHEMATIC Executed record

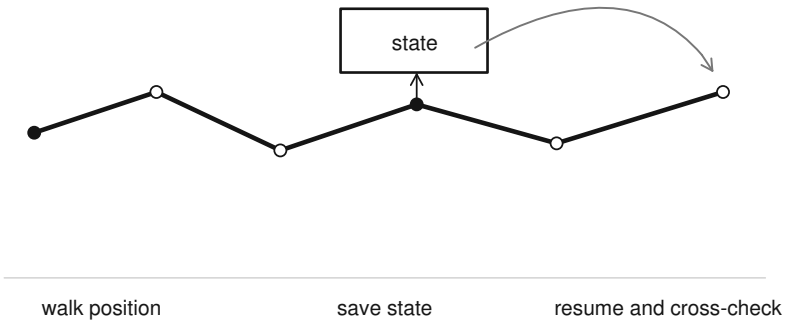
Incremental public-point verification



Finding: Same scalar sequence verified using cached point increments; 900 comparison cases and paired benchmarks passed.

M012 / SCHEMATIC Implemented

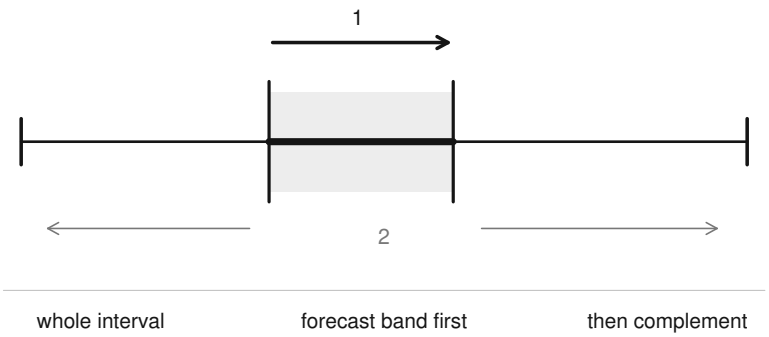
Checkpoint resume and scalar cross-checks



Finding: Live optimization resumes the exact saved position and compares against full scalar multiplication every 10,000 entries.

M013 / SCHEMATIC Executed record

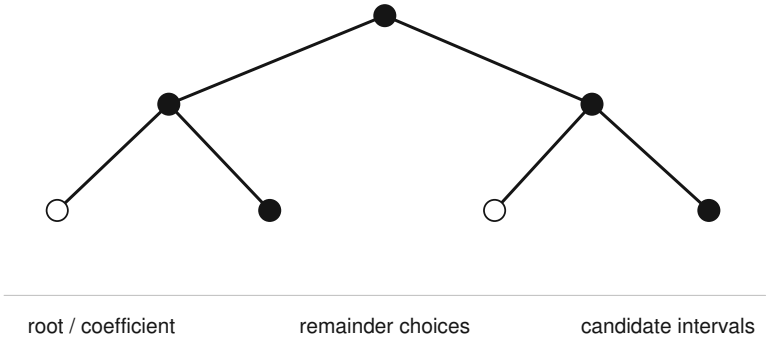
FUSED priority-band traversal with fallback



Finding: Fixed-budget test retained complementary regions; 286,720 checks across both methods found no keys.

M014 / SCHEMATIC Executed record

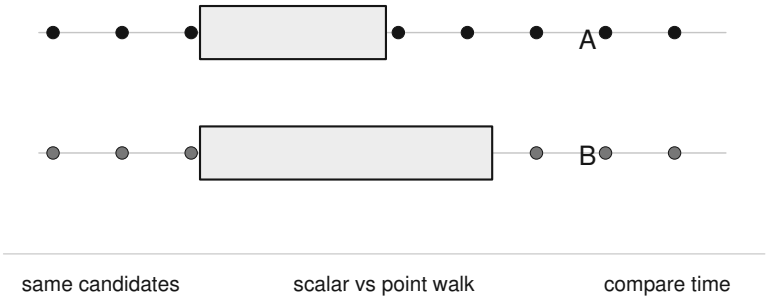
Advanced selective FUSED/Delta2 adapter



Finding: Selected 12,515 hypothesis shards across puzzles 30-64; only 1/35 known targets was contained.

M015 / SCHEMATIC Executed record

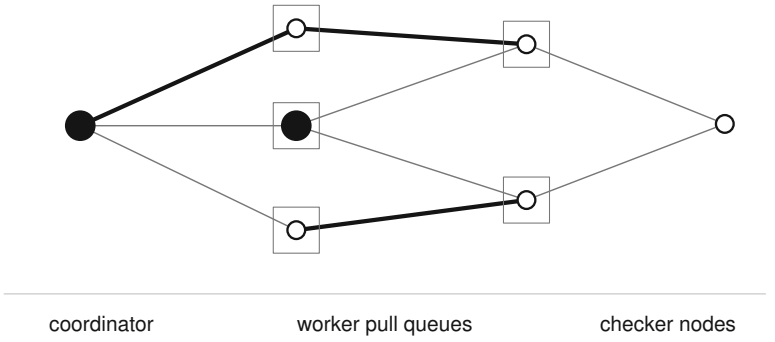
CPU scalar versus point-walk benchmarking



Finding: Measured acceleration is computational, not evidence of a more likely candidate distribution.

M016 / SCHEMATIC Implemented

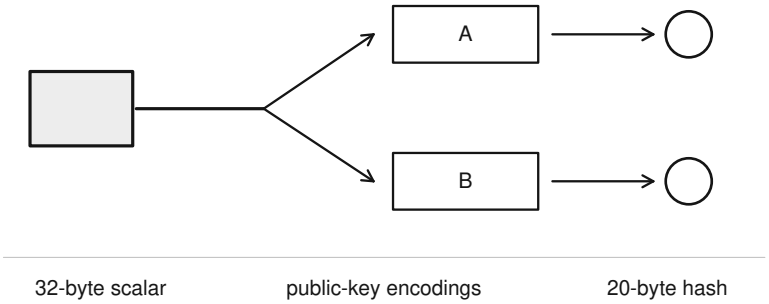
Mock distributed key-generation network



Finding: Pull-based process/thread workers and checker queues exercise scheduling on bounded mock work; not a full puzzle solve.

M017 / SCHEMATIC Executed record

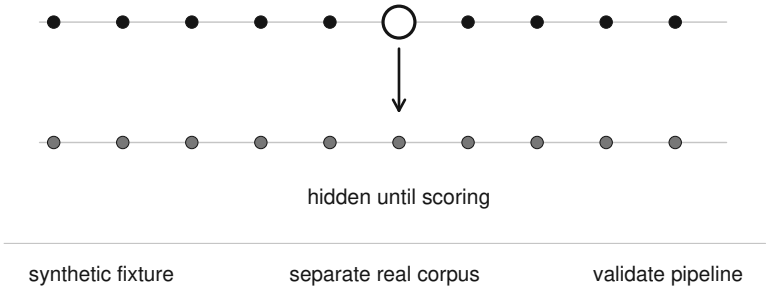
32-byte padding and dual-format P2PKH checks



Finding: Left-padded scalar inputs, compressed/uncompressed public keys and known scalar-one address vectors guard verifier correctness.

M018 / SCHEMATIC Executed record

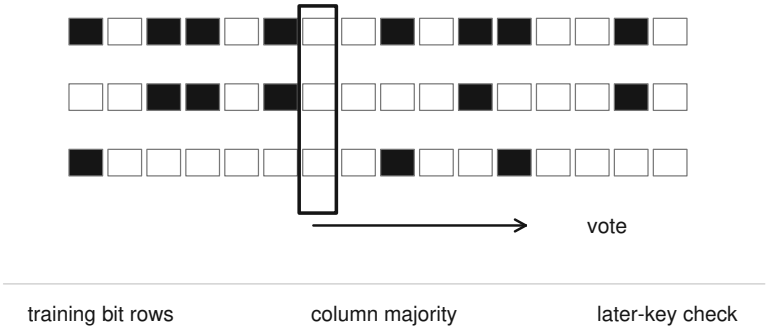
Synthetic fixtures versus disclosed puzzle data



Finding: PEC37 stores synthetic 1-160 chains separately and compares them with actual disclosed keys. Synthetic keys are not puzzle recoveries.

M019 / SCHEMATIC Executed record

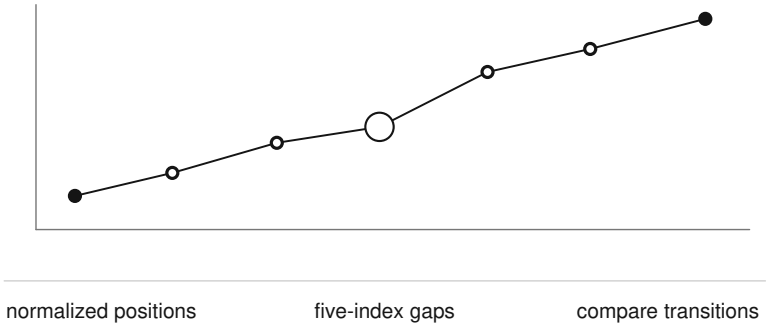
Empirical bit-vote and position forecasting



Finding: Retrospective generator-forensics tests fractional-bit votes, exact regressions and baseline predictors on disclosed keys.

M020 / SCHEMATIC Executed record

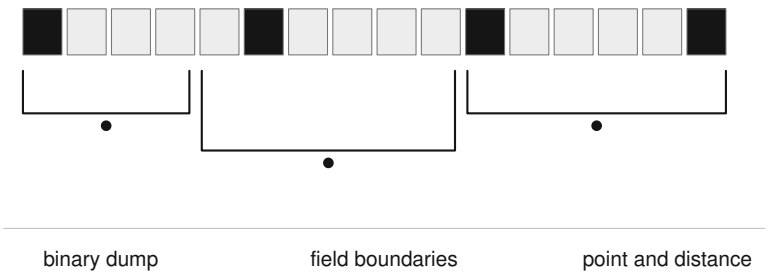
Five-index scale/transition analysis



Finding: Normalized positions, lag correlations, Hamming similarity and held-out transition forecasts are recorded; no new key is recovered.

M021 / SCHEMATIC Executed record

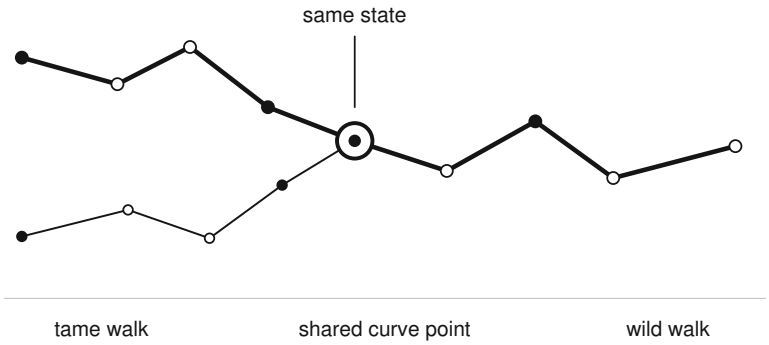
Binary distinguished-point parsing



Finding: Extract curve x-coordinates, candidate metadata and distance fields from the supplied binary. Interpretation of metadata remains a hypothesis.

M022 / SCHEMATIC Implemented

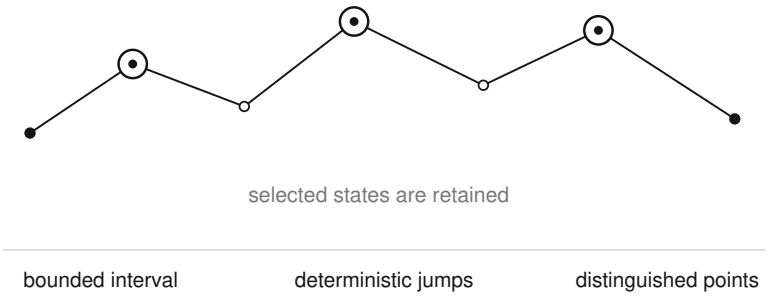
Tame/wild collision recovery



Finding: Same-point opposite-herd distances yield a scalar when the walk origins and distances are valid. No real target collision is established here.

M023 / SCHEMATIC Implemented

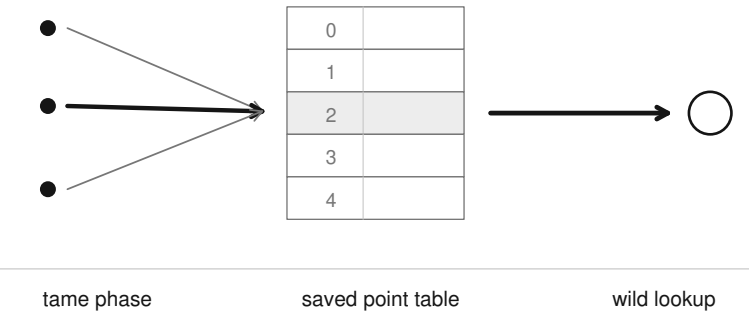
Single-process bounded Pollard kangaroo



Finding: Standard interval discrete-log implementation for targets with known public points.

M024 / SCHEMATIC Implemented

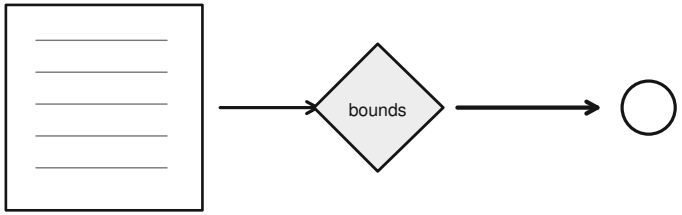
Separated tame and wild phases



Finding: Persist a tame distinguished-point table and search from the target in a separate phase.

M025 / SCHEMATIC Implemented

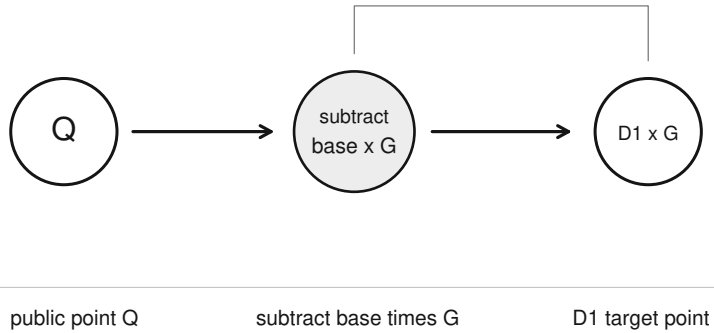
JeanLucPons / RCKangaroo export



Finding: Generate solver configuration files for public-key interval targets; configuration existence is not execution evidence.

M026 / SCHEMATIC Implemented

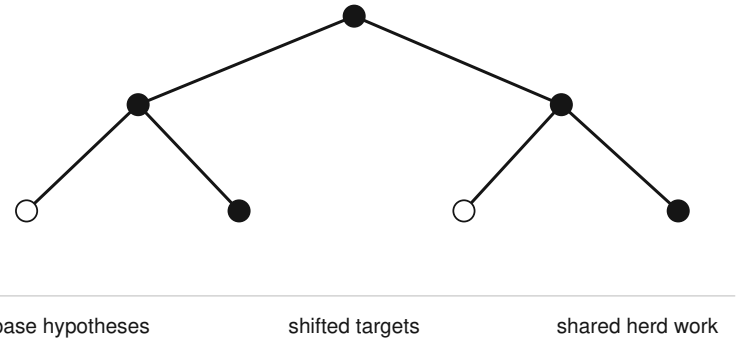
Direct D1 interval kangaroo



Finding: Subtract a hypothesized scalar base and search the D1 leaf; useful only if the hypothesized coordinates are correct.

M027 / SCHEMATIC Audited/corrected

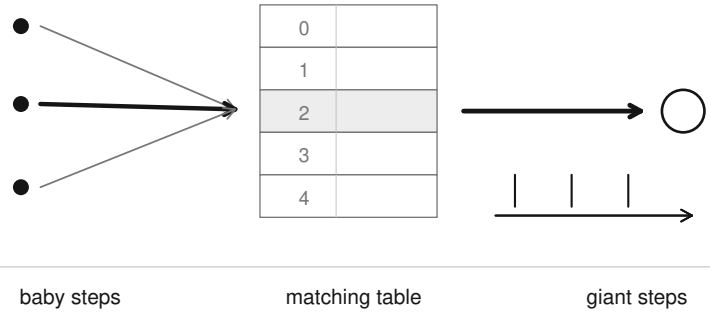
Wrapped CT multi-target kangaroo



Finding: Historical small-span speed claims fail on large required spans; later audits reject unconditional recovery-cost reductions.

M028 / SCHEMATIC Executed record

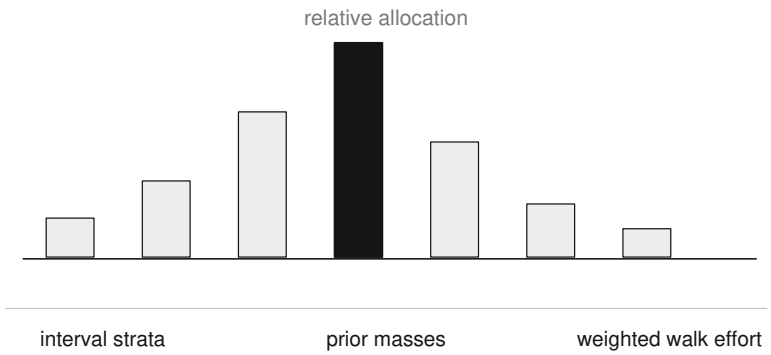
CPU banded baby-step/giant-step D1 hunts



Finding: Bounded D1-band runs are documented; CPU tasks on GPU-equipped pods must not be described as GPU kernel execution.

M029 / SCHEMATIC Implemented

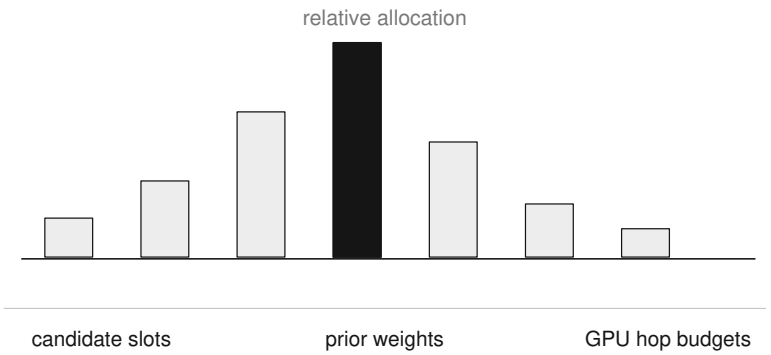
Prior-weighted stratified kangaroo



Finding: Reweights interval effort when a genuine prior exists; flat priors reduce to ordinary allocation. Prior quality is unestablished for the unknown puzzle.

M030 / SCHEMATIC Implemented

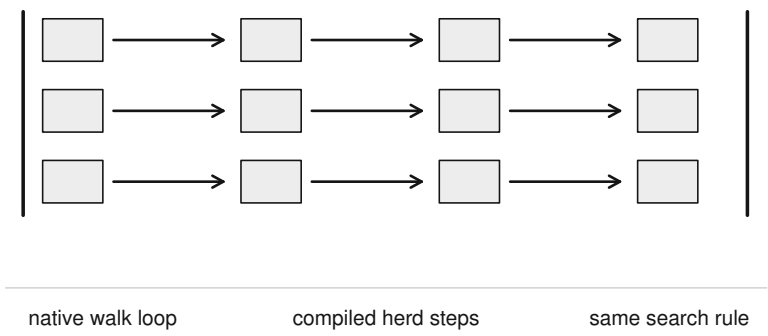
Prior-weighted GPU slot budgets



Finding: Allocate existing slot budgets using FUSED distance or empirical D1 histograms without changing slot geometry.

M031 / SCHEMATIC Implemented

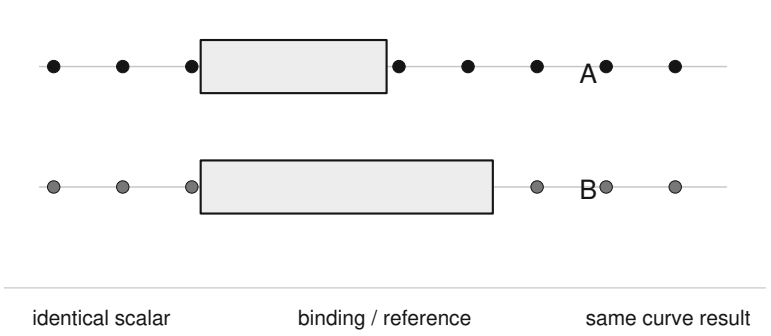
Rust native herd stepping



Finding: Native backend reduces interpreter overhead while preserving interval-search semantics.

M032 / SCHEMATIC Implemented

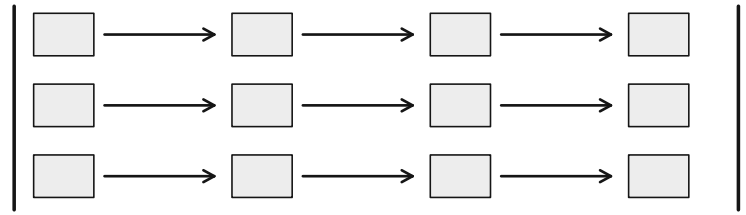
Coincurve/libsecp256k1 fast backend



Finding: Accelerated point arithmetic replaces pure Python operations; it does not reduce unknown key entropy.

M033 / SCHEMATIC Implementation/stub mix

GPU modular herd/D1 backends

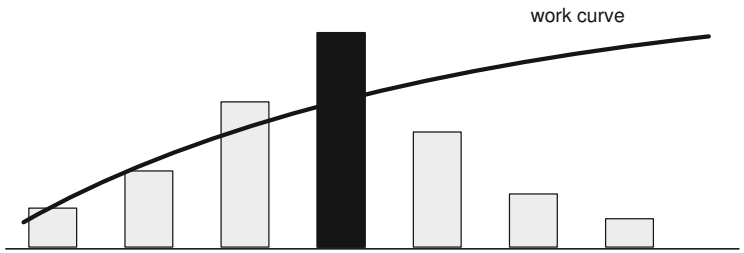


device work groups | parallel walk lanes | implementation required

Finding: Backend registries and CUDA files exist, but readiness and actual execution differ by module. Do not infer a complete device implementation from a registry.

M034 / SCHEMATIC Executed record

GPU throughput and cost calibration

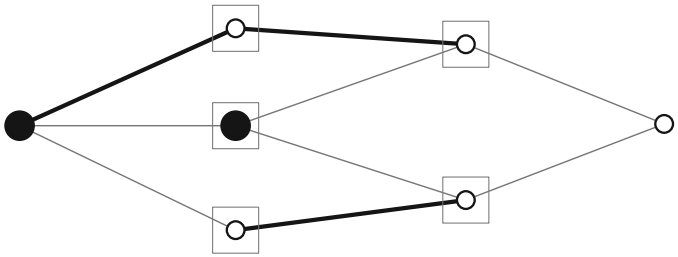


measured throughput | operation budget | time and cost

Finding: Parse measured solver logs separately from hardcoded estimates; historical prices and throughput are not current quotations.

M035 / SCHEMATIC Implemented

RunPod/Vultr sharding and orchestration

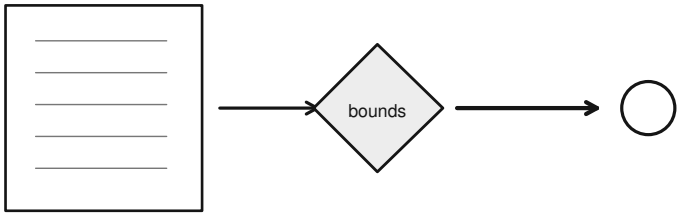


job manifest | remote shards | collected checkpoints

Finding: Deployment runbooks, work-unit manifests and recovery scripts coordinate bounded jobs; availability is distinct from a successful search.

M036 / SCHEMATIC Implemented

Bounded ECDLP certificates and slot gates

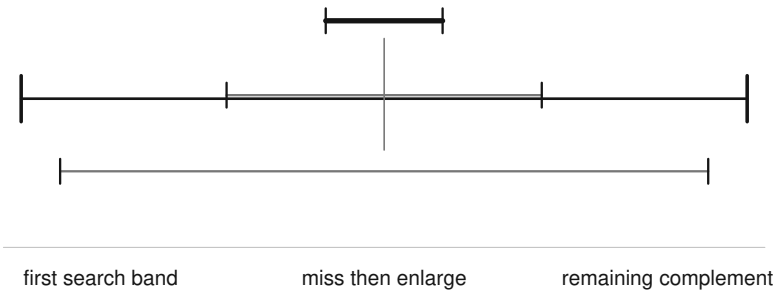


declared bounds | target / provenance | result verification

Finding: Certifies declared bounds and evidence metadata; a certificate is not proof that the target lies in a guessed slot.

M037 / SCHEMATIC Implemented

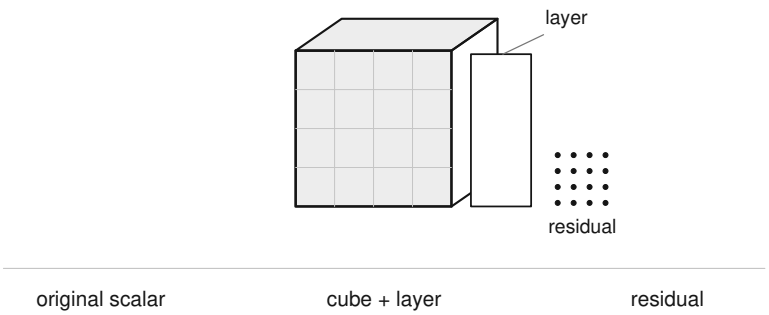
Miss ladder and complement widening



Finding: Follow a failed narrow search with complement/full-range fallbacks, rather than treating heuristic misses as exclusions.

M038 / SCHEMATIC Audited/corrected

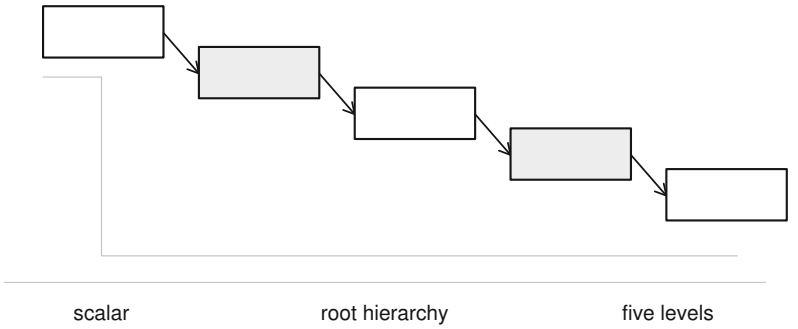
Cubic scalar decomposition



Finding: Exact reconstruction of known integers works; the representation itself supplies no unknown digits or entropy reduction.

M039 / SCHEMATIC Implemented

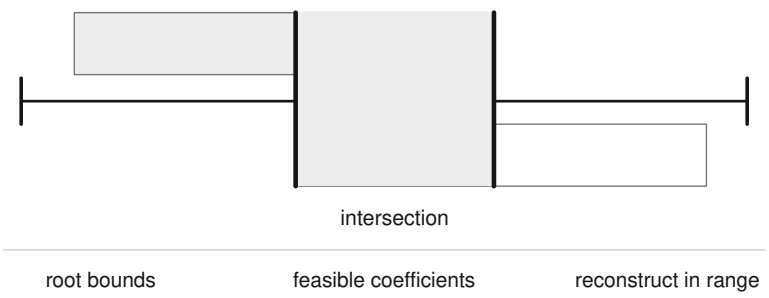
Five-level R1/R2/S3/S4/S5 tower



Finding: Recursive roots and coefficients organize scalar coordinates; high levels often depend only on advertised bit width.

M040 / SCHEMATIC Executed record

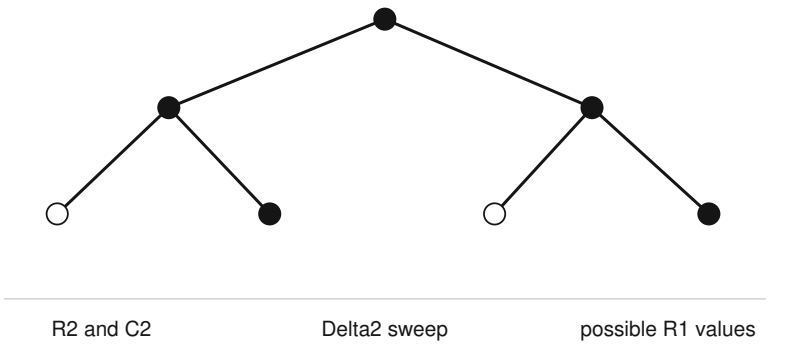
Exact root bounds and coefficient feasibility



Finding: Integer roots, coefficient limits, clipping and reconstruction are tested arithmetically.

M041 / SCHEMATIC Implemented

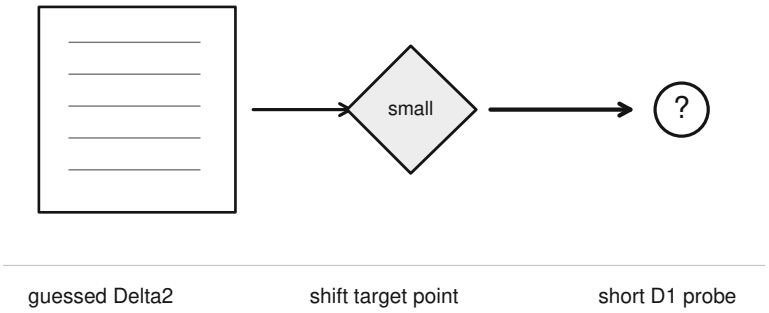
Phase-2 Delta2 enumeration



Finding: Enumerate feasible R1 values within each R2/C2 hypothesis; this is a search coordinate, not an independently observed point.

M042 / SCHEMATIC Implemented

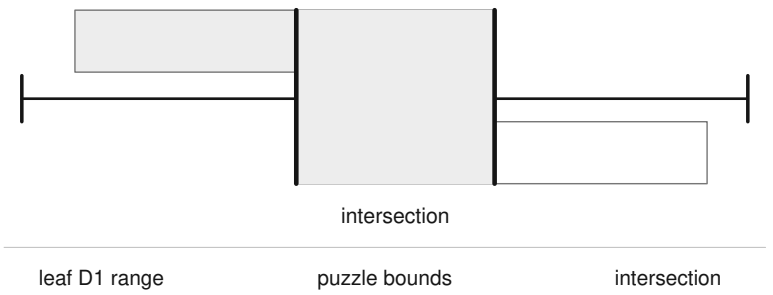
Phase-2 public-key-guided light probes



Finding: Construct D1 targets from Delta2 guesses and run bounded probes; no independent Delta2 oracle is obtained.

M043 / SCHEMATIC Executed record

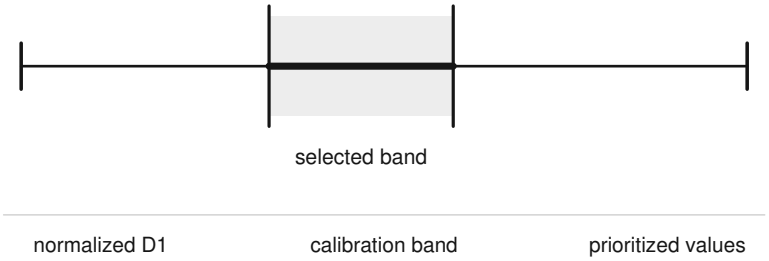
Puzzle-bound clipping of D1



Finding: Historical audit records no useful narrowing for its high-index sample; advertised puzzle bounds are too broad at this leaf scale.

M044 / SCHEMATIC Executed record

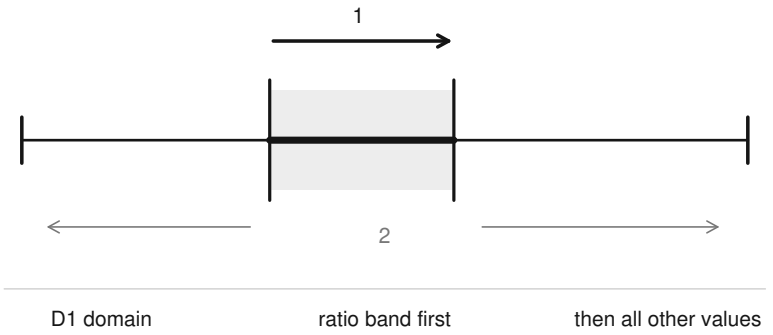
D1 high-index ratio bands



Finding: Calibrated narrow bands reduce width conditionally but miss held-out keys; their complement is required for complete support.

M045 / SCHEMATIC Implemented

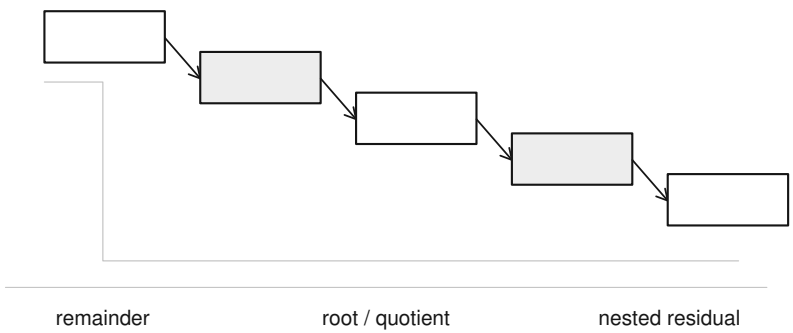
Two-phase D1 band/complement policy



Finding: Prioritize a calibrated band and search its remainder after a miss. Union coverage restores full range rather than reducing entropy.

M046 / SCHEMATIC Executed record

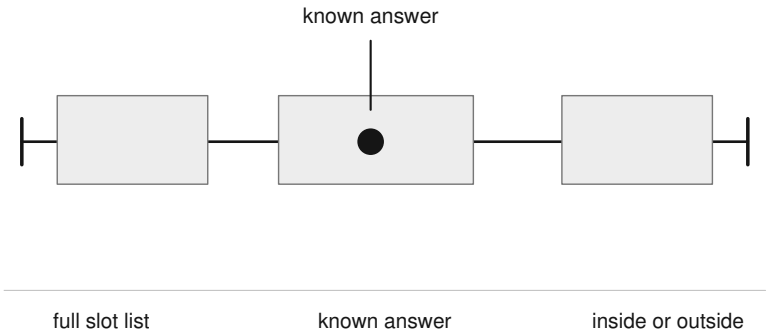
Recursive remainder/sub-digit decomposition



Finding: Deeper splits were compared with random controls; apparent digit biases can arise from the map itself.

M047 / SCHEMATIC Executed record

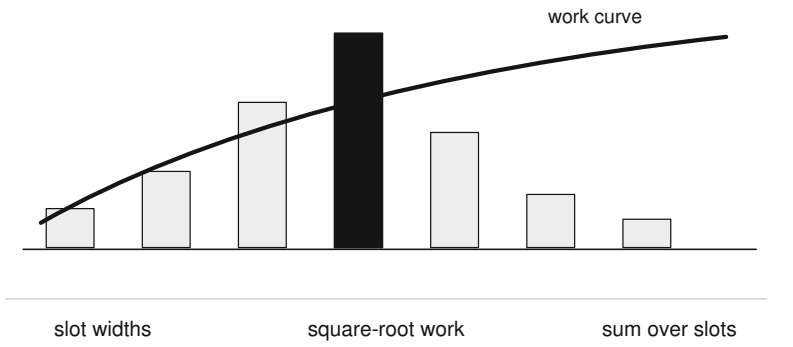
All-known-key slot containment



Finding: Check whether a generated complete slot includes a published answer; containment is not independent discovery of that answer.

M048 / SCHEMATIC Implemented

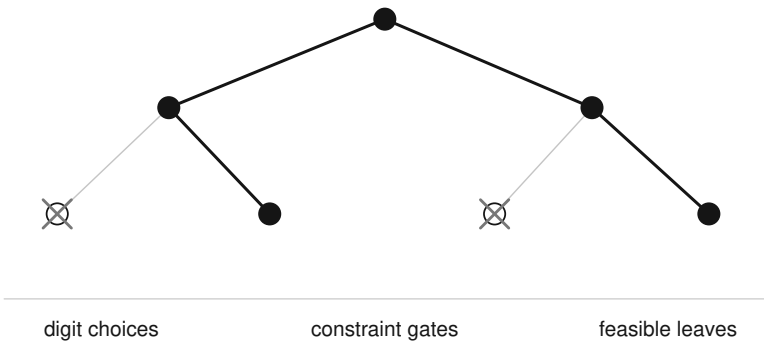
Post-slot square-root cost scaling



Finding: Count leaf cost conditional on a correct slot; audits warn against omitting the number of unknown slots.

M049 / SCHEMATIC Executed record

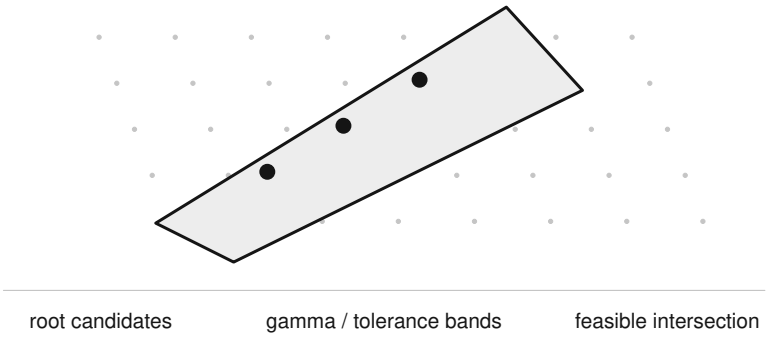
Constraint-satisfaction slot construction



Finding: Exact feasibility graph with fixed, MRV and degree orderings; pruning invalid tuples does not predict the valid target tuple.

M050 / SCHEMATIC Executed record

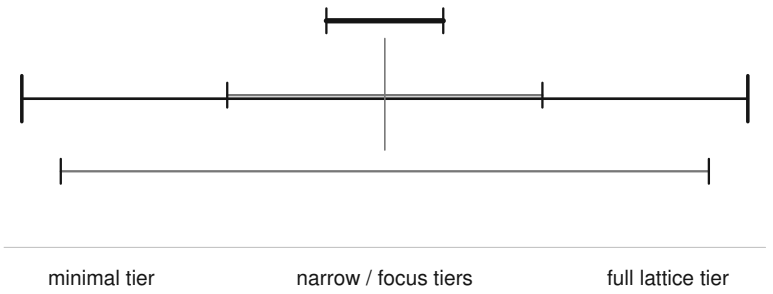
Gamma-band and lattice narrowing presets



Finding: Compare full lattice, gamma restrictions and FUSED intersections; stronger cuts reduce measured target coverage.

M051 / SCHEMATIC Executed record

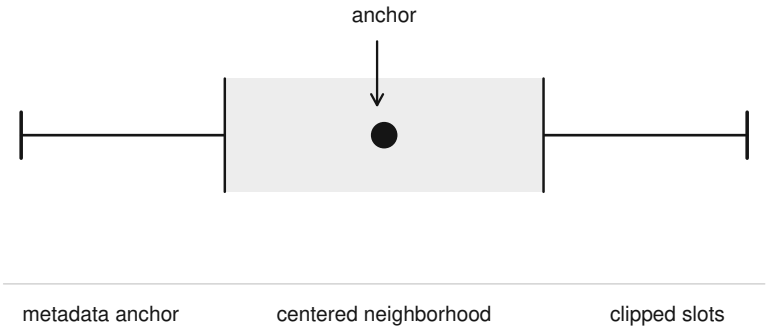
Tiered minimal/narrow/focus/lattice exports



Finding: Saved manifests can exclude the known calibration target; a valid export is not proof of target inclusion.

M052 / SCHEMATIC Implemented

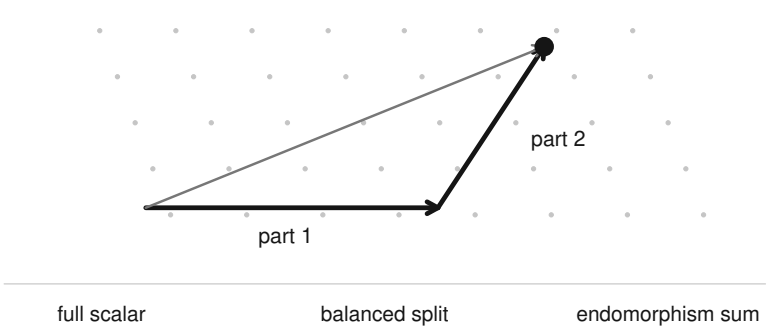
Anchor/TRAILING-centered slot construction



Finding: Metadata-centered guesses are explicit hypotheses and require cryptographic verification.

M053 / SCHEMATIC Executed record

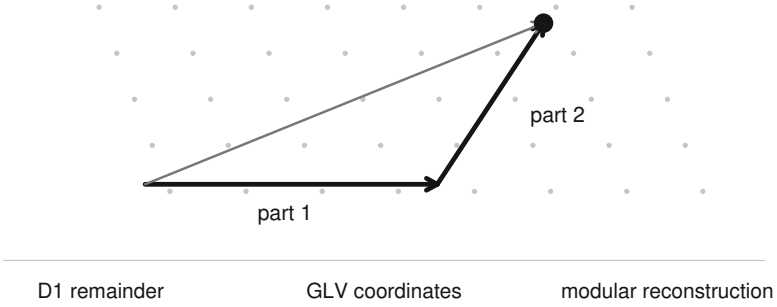
GLV scalar decomposition and endomorphism



Finding: Correct arithmetic acceleration checked on known points; often degenerate for small scalars and not a generic entropy cut.

M054 / SCHEMATIC Audited/corrected

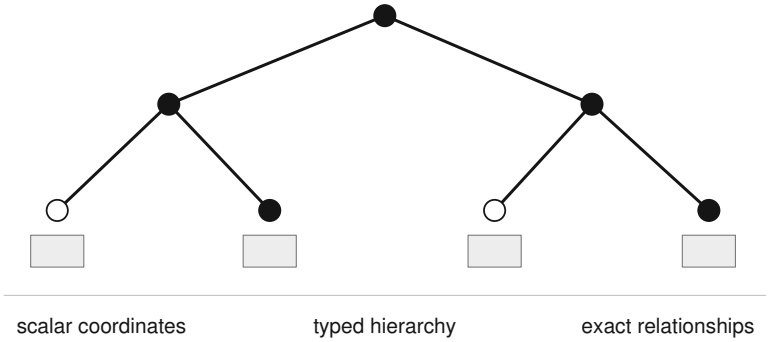
GLV splitting specifically on D1



Finding: Leaf scalars far below group order do not obtain the advertised balanced split; no free narrowing follows.

M055 / SCHEMATIC Implemented

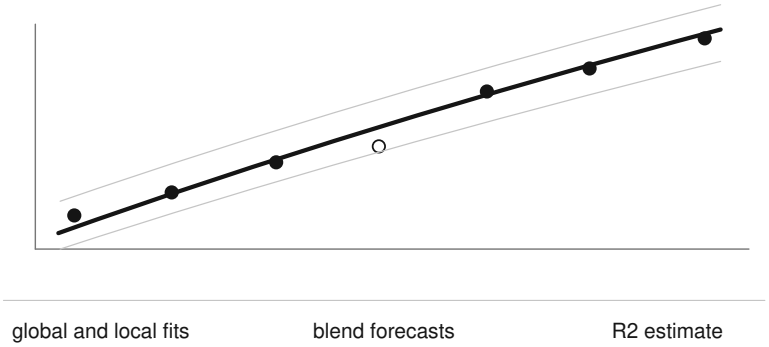
Exact value atlas and hierarchy visualizations



Finding: Typed exports, scientific plots and offline views expose values and bounds; visualization itself is not a predictor.

M056 / SCHEMATIC Executed record

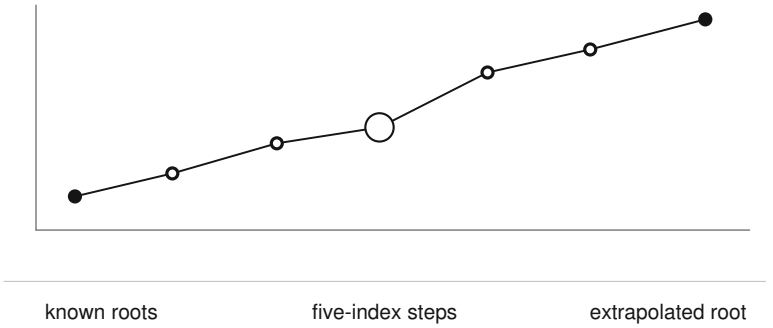
FUSED global/local R2 regression



Finding: Historical blends forecast ninth roots; narrow transformed error must be converted to actual scalar coverage.

M057 / SCHEMATIC Executed record

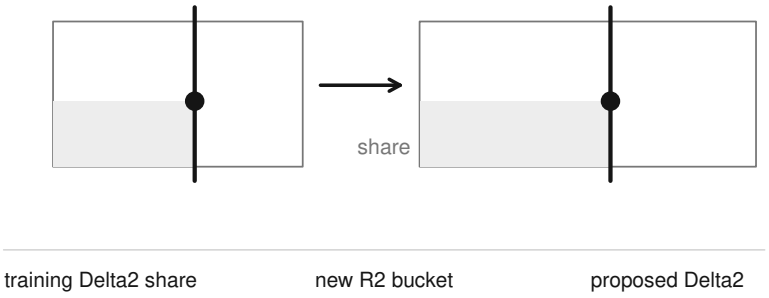
Gap-five sequence extrapolation



Finding: Forecast R2 or remainder ratios from every-fifth-puzzle sequences; tested against forward-only and uniform controls.

M058 / SCHEMATIC Audited/corrected

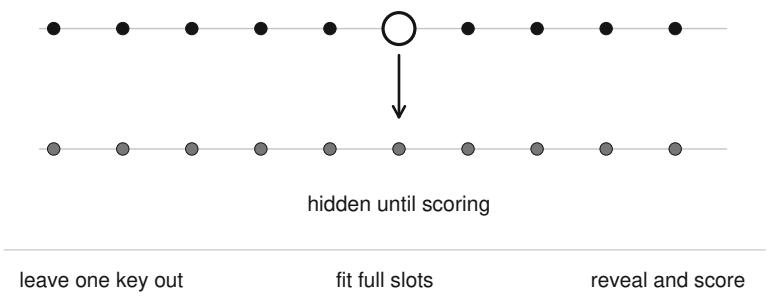
Strategic Delta2 ratio hints



Finding: Target-in-calibration leakage explained the old 7/7 result; corrected blind high-index results were 0/7.

M059 / SCHEMATIC Executed record

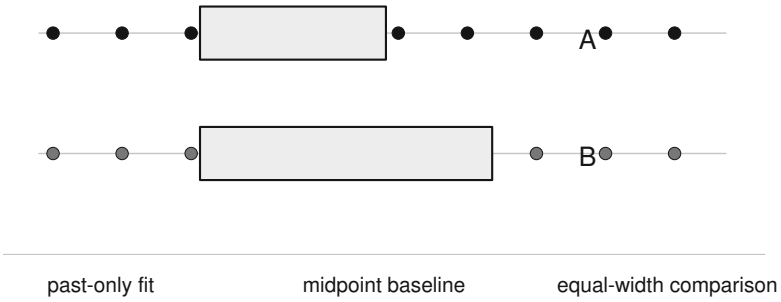
Leave-one-out full-slot backtest



Finding: Corrected full-slot requirements fail in the documented seven high-index holdouts, despite coarse R2 coverage.

M060 / SCHEMATIC Executed record

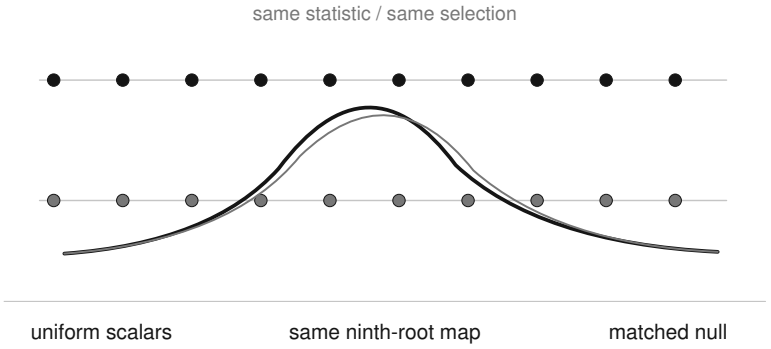
Past-only FUSED versus midpoint



Finding: Same-band comparisons show that transformed midpoint baselines can rival or beat apparent R2 coverage.

M061 / SCHEMATIC Audited/corrected

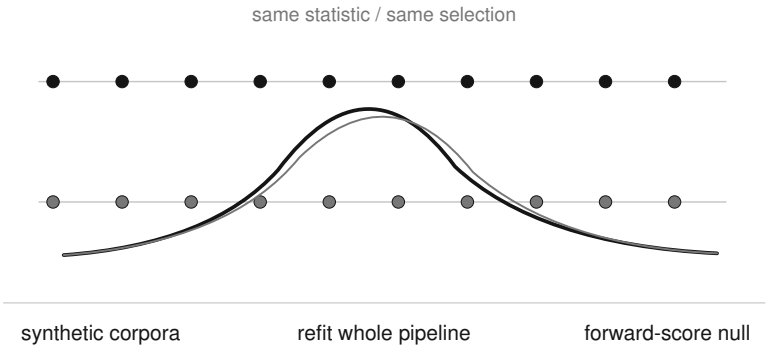
Correct uniform null for ninth-root clustering



Finding: Uniform scalars naturally cluster after ninth-root transformation; an earlier overly broad null standard deviation was corrected.

M062 / SCHEMATIC Executed record

Forward-only synthetic-corpus null



Finding: Repeat the predictor-fitting process on uniform synthetic keys and count exact scalar coverage.

M063 / SCHEMATIC Executed record

Family-wise forward R2 grid



Finding: Test selection across FUSED, gap-five, midpoint, cutoffs and window widths with family-wise controls.

M064 / SCHEMATIC Executed record

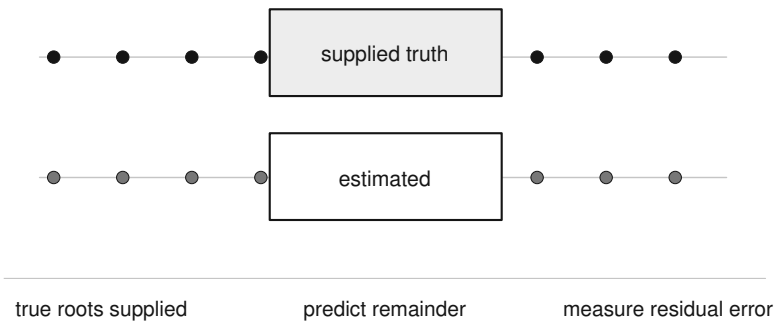
Contiguous-slice selection adjustment



Finding: Compare best-looking index slices with similarly selected synthetic controls; guards against picking favorable subsequences.

M065 / SCHEMATIC Executed record

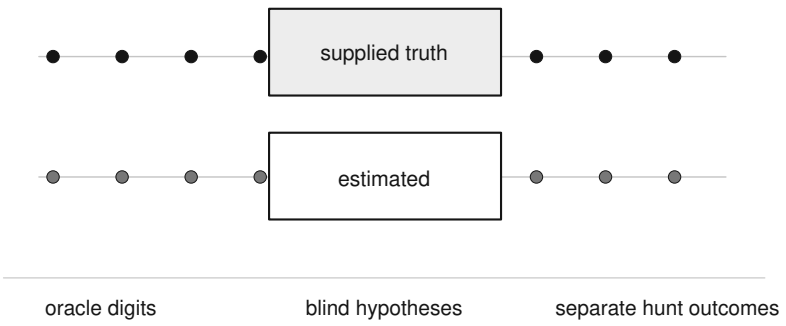
Remainder precision granted true roots



Finding: Measures exact offset-prediction errors even when root coordinates are supplied; coarse root prediction does not supply leaf precision.

M066 / SCHEMATIC Audited/corrected

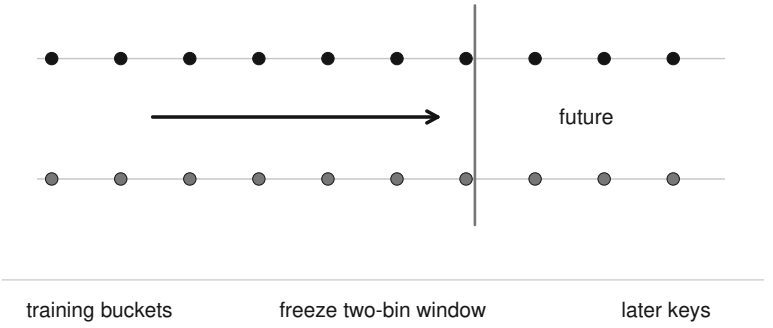
Oracle-versus-blind hunt comparison



Finding: Distinguishes provided true coordinates, key containment, and actual address/public-point discovery.

M067 / SCHEMATIC Executed record

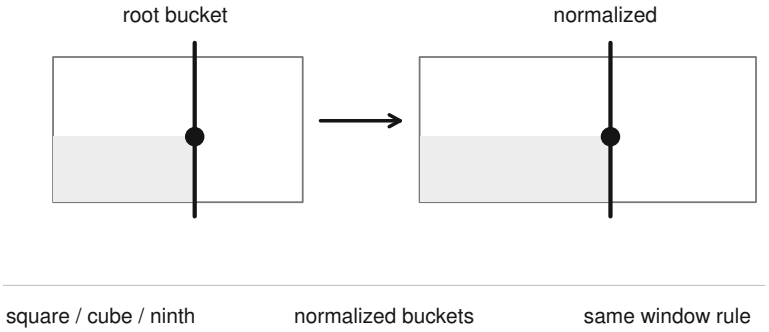
Frozen ninth-root remainder window



Finding: Recent test trained on 17-32 and covered 7/32 evaluation keys; uncorrected p about 0.116, not convincing evidence.

M068 / SCHEMATIC Executed record

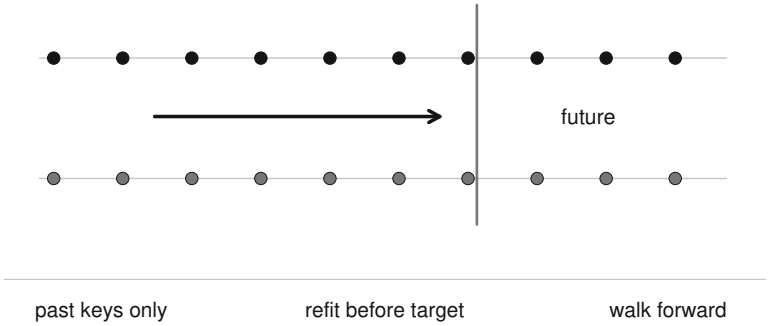
Square/cube/ninth-root window family



Finding: Sixteen frozen models across four nominal widths; none significant after multiple-model correction.

M069 / SCHEMATIC Executed record

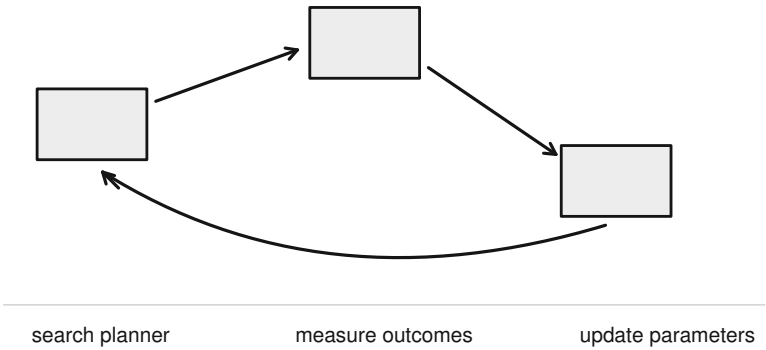
Walk-forward adaptive remainder windows



Finding: Refit only on lower indices; final evidence scores did not exceed the family threshold.

M070 / SCHEMATIC Implemented

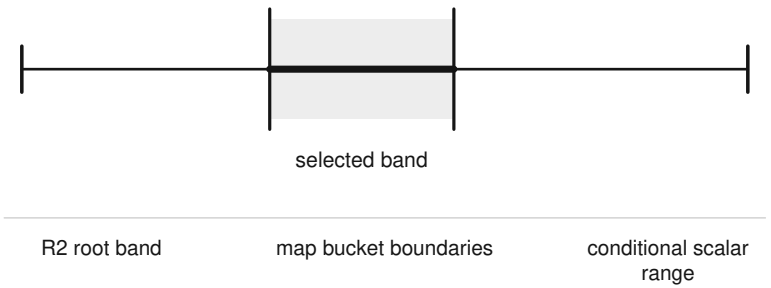
Self-improving CTheory parameter loop



Finding: Measure/analyze/improve loop tunes hint budgets, R2 width and slots; usefulness is limited by honest target inclusion.

M071 / SCHEMATIC Executed record

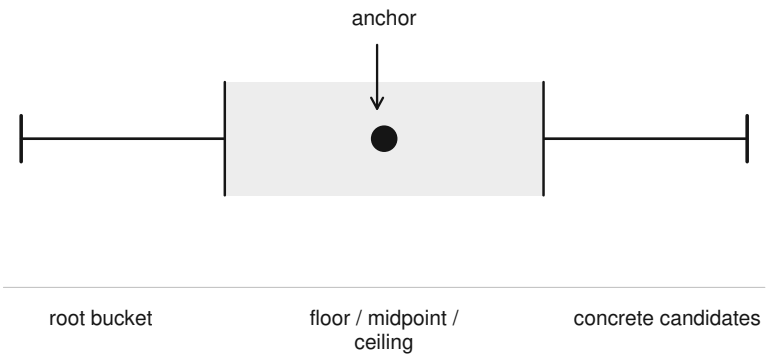
R2 intervals as conditional search prioritization



Finding: Root forecast bands map to exact scalar intervals; reports distinguish conditional narrowing from guaranteed coverage and statistical advantage.

M072 / SCHEMATIC Executed record

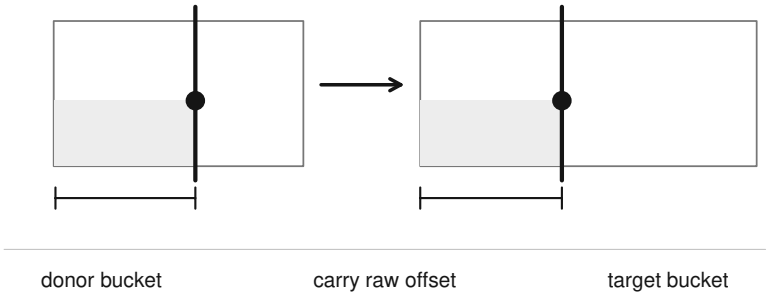
Root floor/midpoint/ceiling candidate anchors



Finding: Fixed root-band and root-difference candidate families checked against known and unknown targets.

M073 / SCHEMATIC Executed record

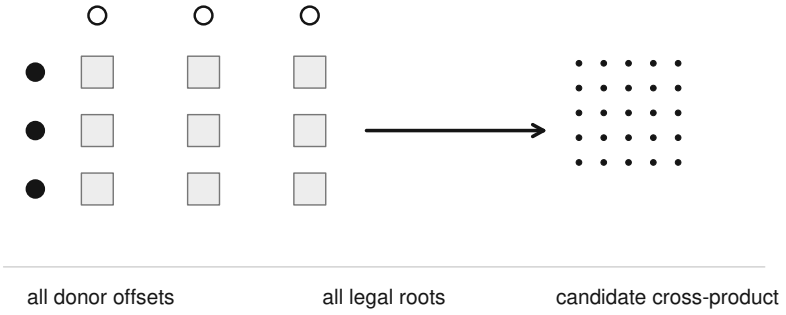
Nearest-donor raw remainder carry



Finding: Transfer a disclosed key remainder to target root hypotheses; saved unknown-target candidate set contains no matches.

M074 / SCHEMATIC Executed record

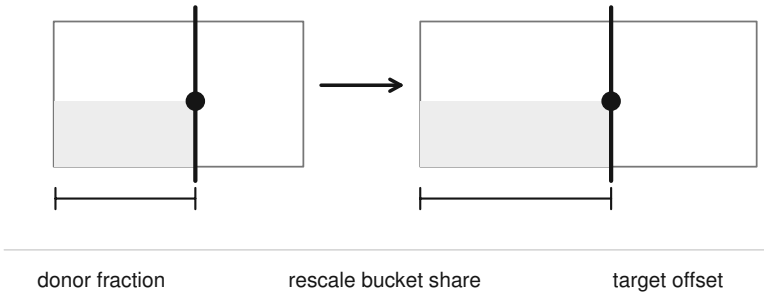
All-donor carry over every legal ninth root



Finding: Saved exhaustive-within-family report records 35,982,885 candidate checks and zero target matches.

M075 / SCHEMATIC Executed record

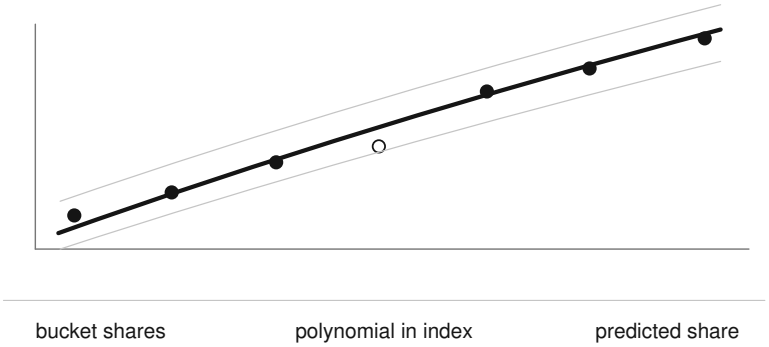
Transferred within-bucket share



Finding: Reuse disclosed normalized root-bucket shares at legal target roots; tests do not cover arbitrary offset functions.

M076 / SCHEMATIC Executed record

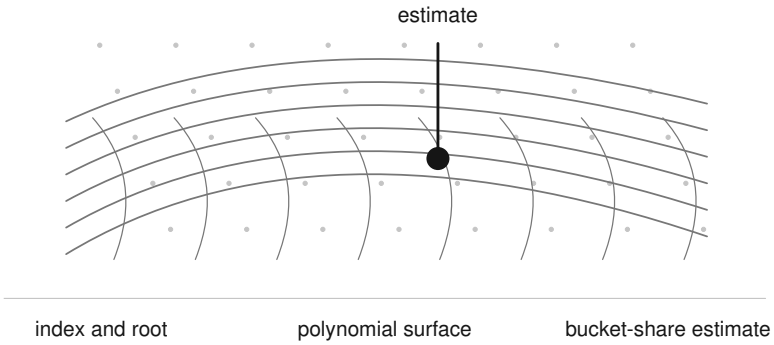
Smooth polynomial share in puzzle index



Finding: Degrees 1-4 predict bucket shares; saved per-degree unknown-target tests report no matches.

M077 / SCHEMATIC Executed record

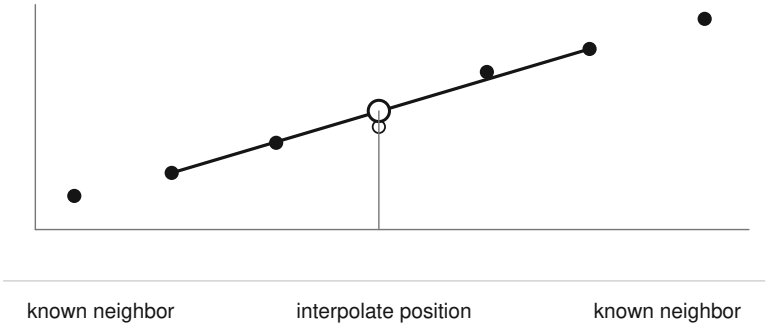
Two-dimensional index/root share surface



Finding: Fit share as a low-degree surface of index and root, then verify resulting candidates; no saved unknown-target hits.

M078 / SCHEMATIC Executed record

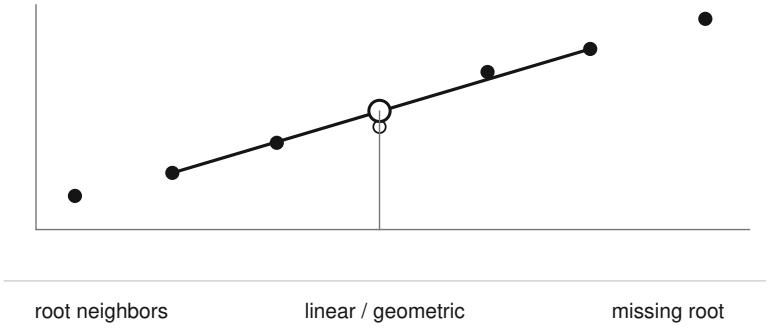
Gap-fill normalized-position interpolation



Finding: Interpolate/extrapolate between disclosed neighboring puzzles and verify candidates; no recorded matches.

M079 / SCHEMATIC Executed record

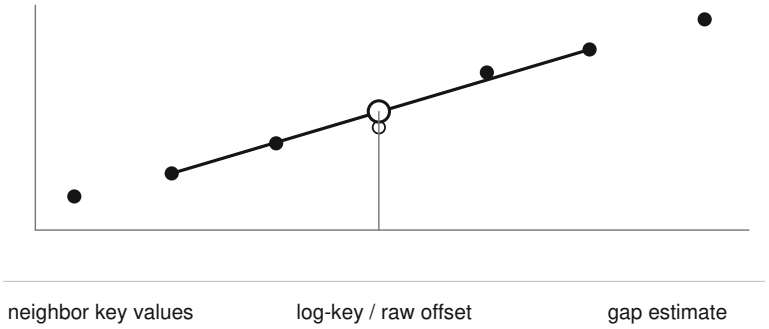
Gap-fill linear/geometric root interpolation



Finding: Interpolate root coordinates and reconstruct candidate powers; tested candidates did not recover unknown keys.

M080 / SCHEMATIC Executed record

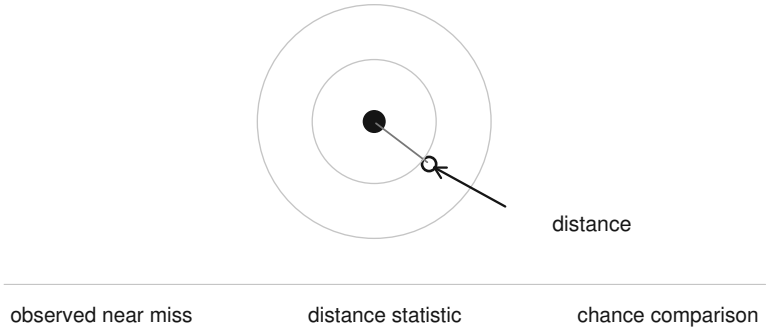
Gap-fill log-key and raw-offset interpolation



Finding: Alternative smoothness assumptions tested against exact address hashes, not judged by numerical proximity alone.

M081 / SCHEMATIC Executed record

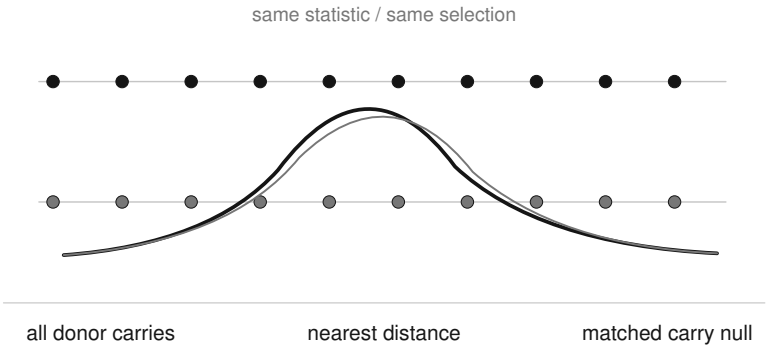
Near-miss anomaly testing



Finding: Compare nearest-candidate distances with uniform targets against the same candidate geometry; closeness alone is not a verified key.

M082 / SCHEMATIC Executed record

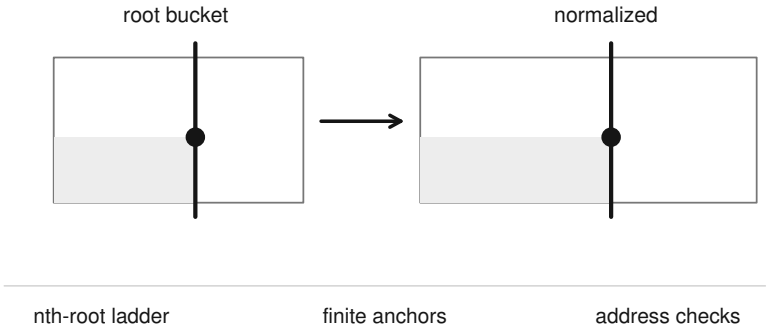
All-donor carry proximity null



Finding: Compare minimum carry distance to uniform controls while excluding the known target from its donor set.

M083 / SCHEMATIC Executed record

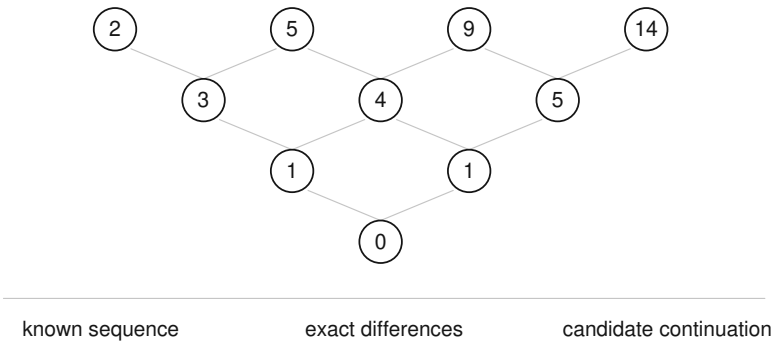
Blind nth-root ladder address tests



Finding: Index-only, leave-one-out and random holdout candidates mostly match only trivial tiny puzzles; no unknown-target matches recorded.

M084 / SCHEMATIC Executed record

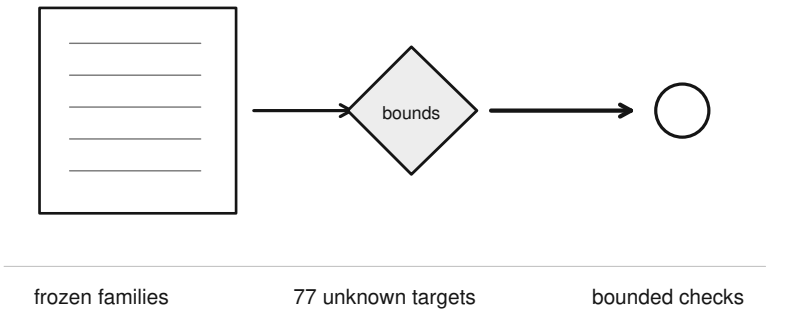
Exact polynomial/difference continuation



Finding: Degree 0-6 models of scalars, range positions and roots failed exact higher-index recovery in the saved tests.

M085 / SCHEMATIC Executed record

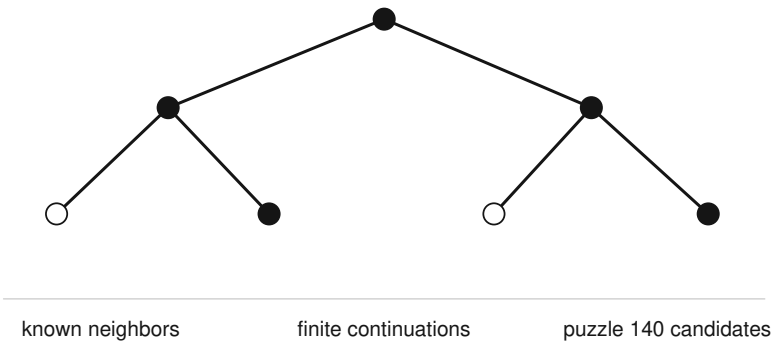
Frozen candidate families on 77 unknown targets



Finding: Local all-unknown report records 2,744 distinct candidate/target checks and no matches.

M086 / SCHEMATIC Executed record

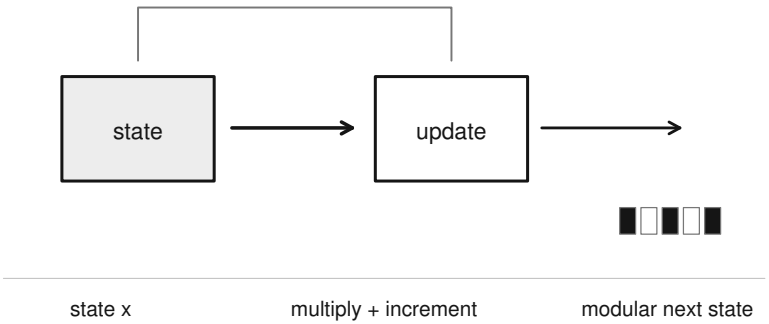
Finite continuation attempts specifically for #140



Finding: Archive leads and binary candidates were verified as bounded hypotheses, not exhaustive interval searches.

M087 / SCHEMATIC Executed record

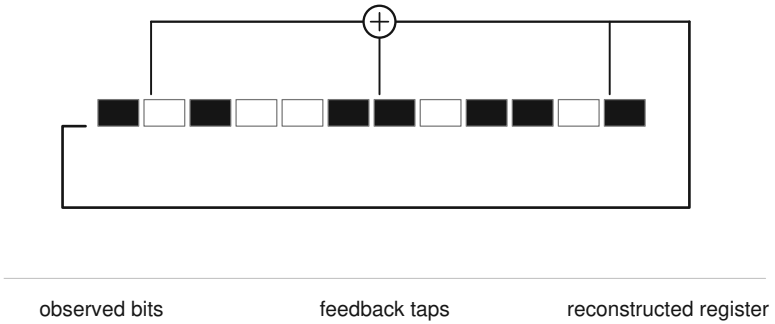
Fixed affine/LCG key recurrence



Finding: Exact low-bit constraints reject the named fixed-parameter recurrence family; hidden state and different generators remain outside scope.

M088 / SCHEMATIC Recorded research

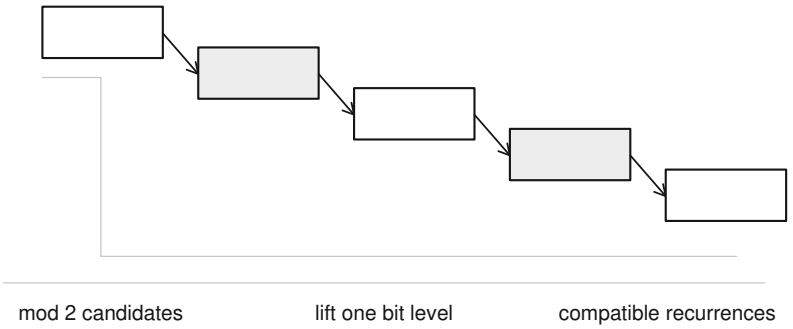
LFSR / Berlekamp-Massey reconstruction



Finding: Historical generator review includes bitstream linear-recurrence tests; no transferable unknown-key generator established.

M089 / SCHEMATIC Recorded research

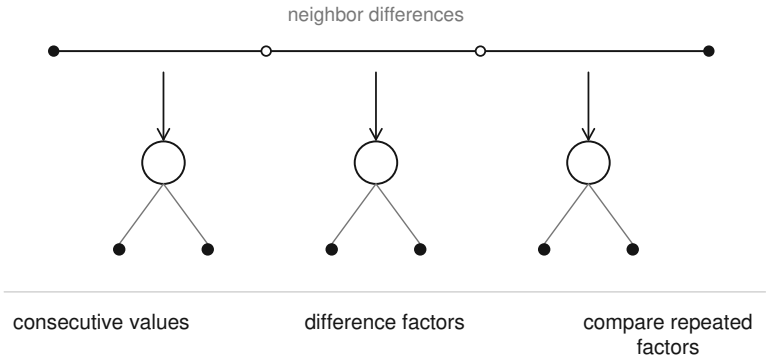
Two-adic/Hensel polynomial recurrence



Finding: Tests of squared-state and modular recurrence hypotheses are archived; finite failures are not impossibility theorems.

M090 / SCHEMATIC Recorded research

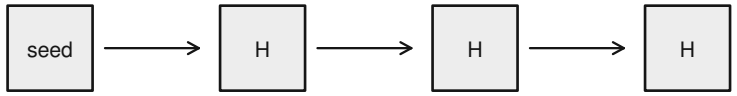
Consecutive-difference factor patterns



Finding: Search common factors or structured differences across published puzzle keys; no validated target predictor recorded.

M091 / SCHEMATIC Executed record

Hash-chain continuation

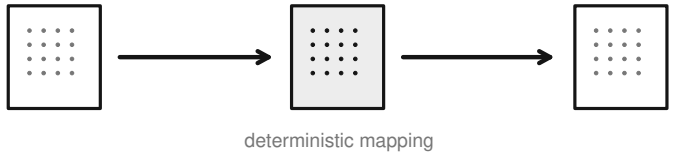


starting value repeated hash next digest

Finding: SHA-256/double-hash variants of previous keys, with endian/length choices and masks, failed named holdout and target checks.

M092 / SCHEMATIC Implemented

Seed-plus-index hash/HMAC families

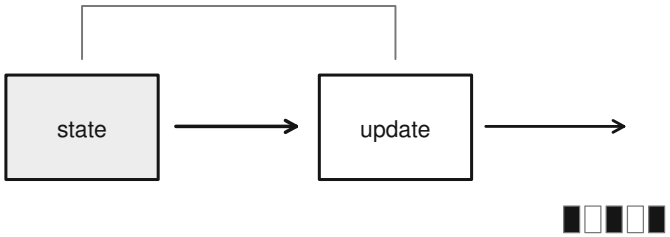


seed and index hash / HMAC candidate output

Finding: Score candidate seeds over different index encodings; requiring significant later-tier matches avoids trivial small-range coincidences.

M093 / SCHEMATIC Implemented

Python MT19937 masked-key replay

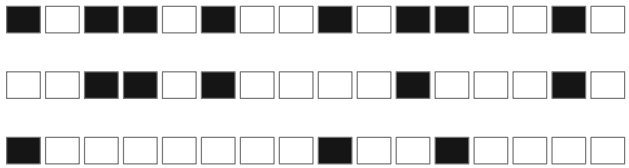


Python MT seed word stream puzzle-width mask

Finding: Test seeded randrange/getrandbits generation schedules against published puzzle keys.

M094 / SCHEMATIC Executed record

Shared unmasked value, prefix and suffix tests

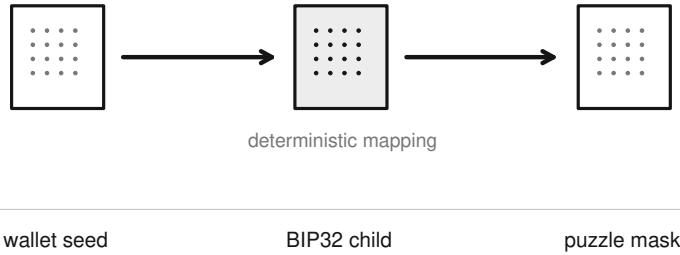


unmasked candidates prefix / suffix overlap shared-value test

Finding: Known-key overlap and random controls reject the tested common-prefix/suffix hypotheses, not every wallet generator.

M095 / SCHEMATIC Executed record

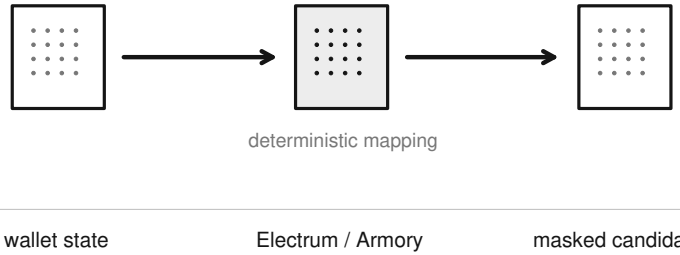
CoinVault/mSIGNA BIP32 plus mask



Finding: Bounded seed/path catalog did not reproduce nontrivial puzzle keys; synthetic known-wallet recovery validates the checker.

M096 / SCHEMATIC Implemented

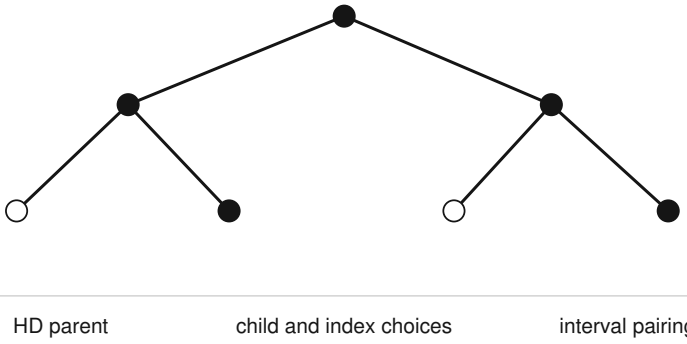
Electrum-v1 and Armory masking models



Finding: Historical deterministic-wallet derivation families and creator-style masks have dedicated test scripts.

M097 / SCHEMATIC Implemented

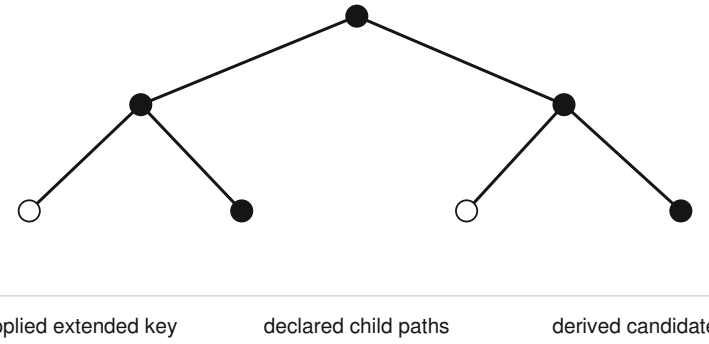
BIP32 child/index interval pairing



Finding: Enumerate path-template index/interval pairs under an explicitly supplied seed; synthetic self-test is separate from creator identification.

M098 / SCHEMATIC Executed record

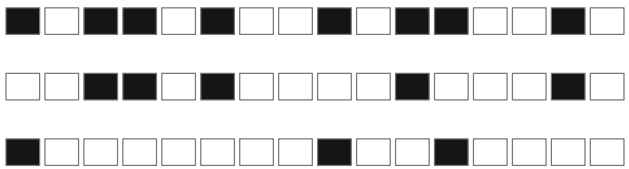
Supplied extended-key derivation leads



Finding: Three unique supplied leads tested over bounded relative paths/index shifts; no reproduction of nontrivial probe keys. Secrets are not included here.

M099 / SCHEMATIC Implemented

Entropy-prefix and full-width constraints

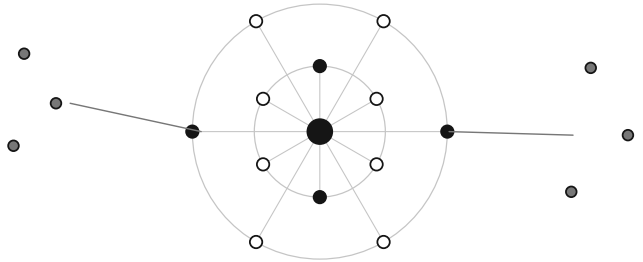


seed constraints | prefix and full width | allowed candidates

Finding: Named candidate scripts examine how scalar masking and source entropy could constrain keys; no generic recovery follows.

M100 / SCHEMATIC Executed record

CLI fingerprint and key clustering

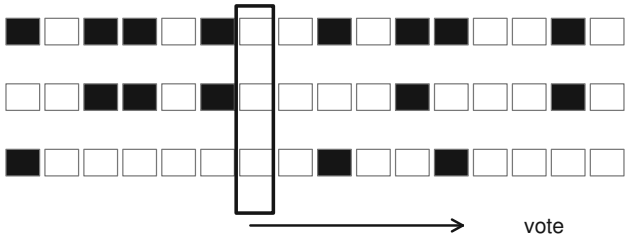


key / CLI records | shared fingerprints | candidate clusters

Finding: Hamming, low-word masks and partial fingerprints organize known coordinates; not established target narrowing.

M101 / SCHEMATIC Executed record

Popcount and aligned-bit distribution tests

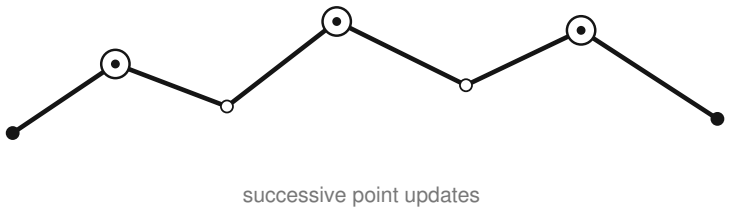


aligned bit rows | popcount / column rates | compare distributions

Finding: Known-key bit densities and overlap are compared with controls; observed appearance is not a scalar equation.

M102 / SCHEMATIC Executed record

Counter/small-affine early public-key relations

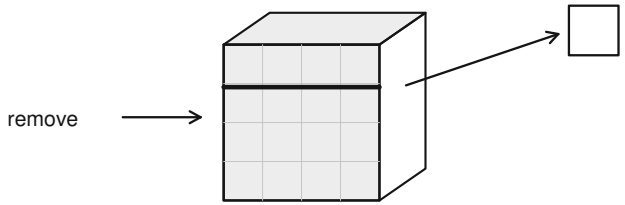


public points | counter / small offsets | test linear relations

Finding: Recorded all-pairs and focused early-key coefficient sweeps found no relation in their finite coefficient/offset bounds.

M103 / SCHEMATIC Executed record

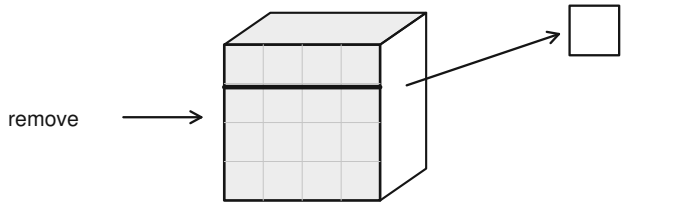
Sparse ORIG decomposition and peel chain



Finding: A constructed sparse scalar peels to tiny known puzzle values; this does not generate the numbered puzzle sequence.

M104 / SCHEMATIC Audited/corrected

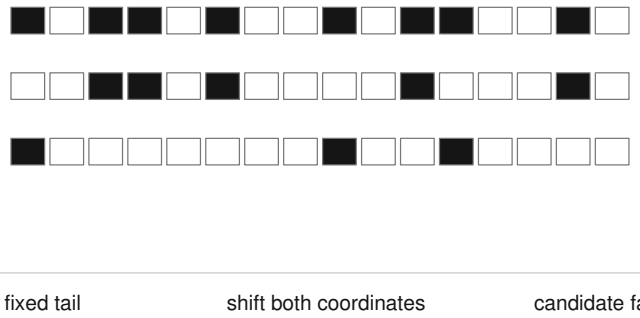
Nested previous-key tail grow/peel



Finding: Distinguishes nested-tail updates from fixed-tail co-shifts; the two agree at one construction step, not generally.

M105 / SCHEMATIC Implemented

Fixed-tail co-shift family



Finding: Exact sparse-key program shifts its frame while preserving the tail; valid program structure is not a Bitcoin predictor.

M106 / SCHEMATIC Executed record

Sparse bit-learning and dial-in grids



Finding: Learn/test mid, gap, bit and frame variants; archived investigation reports poor known-key recall and no unknown address hits.

M107 / SCHEMATIC Executed record

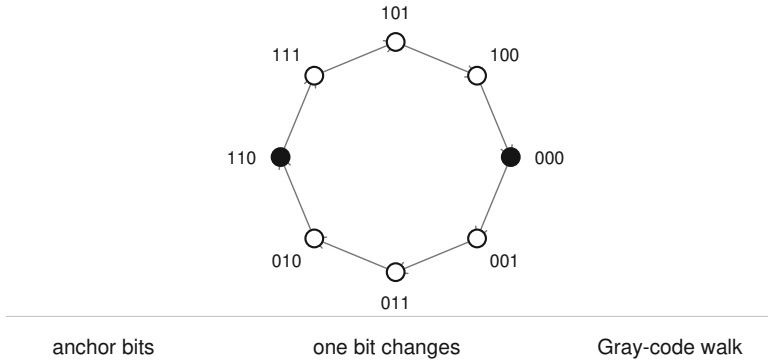
Sparse XOR-tail / delta / Bloom variants



Finding: Alternate sparse generative families were tested without establishing a transferable key-generation rule.

M108 / SCHEMATIC Implemented

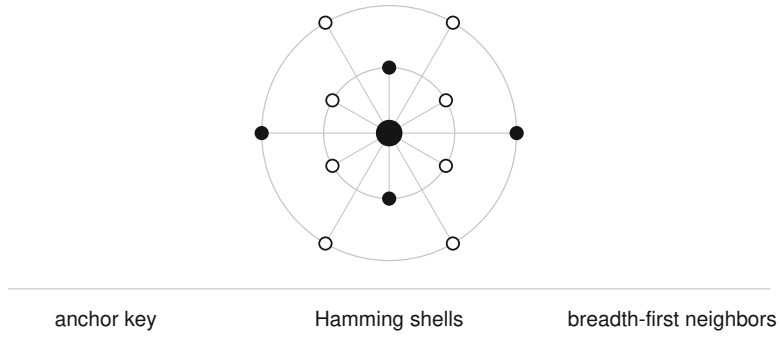
Gray-code range and anchor walks



Finding: Enumerate Gray offsets around range floors or metadata anchors; this changes order rather than entropy.

M109 / SCHEMATIC Implemented

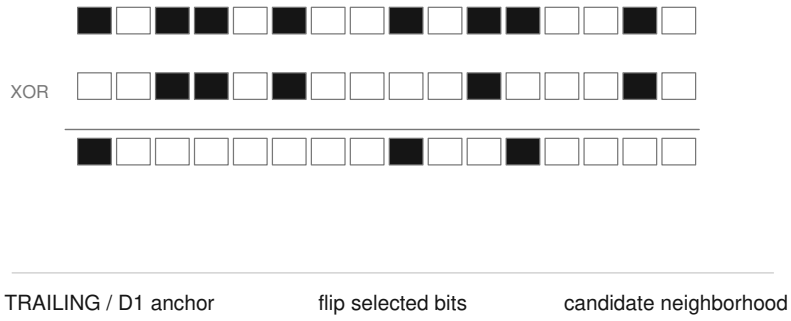
BFS Hamming balls and previous-key XOR



Finding: Check bounded low-Hamming-distance neighborhoods around seeds or prior keys; completeness is limited to that neighborhood.

M110 / SCHEMATIC Implemented

TRAILING-mask and FUSED-D1 bit flips



Finding: Restrict flipped positions or anchor the D1 coordinate; target inclusion is unproven.

M111 / SCHEMATIC Implemented

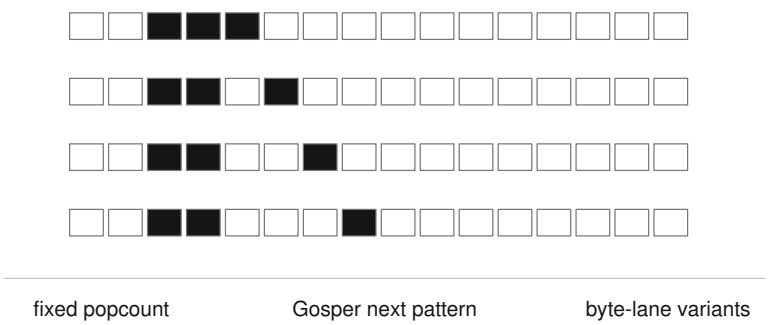
Dual-mask byte/bit and Gray-PDEP walks



Finding: Deposit counter bits into complementary masks, including Gray ordering; these are deterministic traversal families.

M112 / SCHEMATIC Implemented

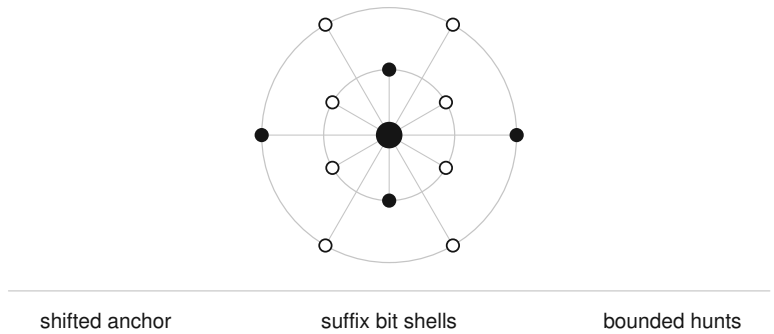
Gosper constant-weight and byte-lane variants



Finding: Visit fixed-weight XOR masks or replace individual bytes; no claim that these neighborhoods contain the target.

M113 / SCHEMATIC Implemented

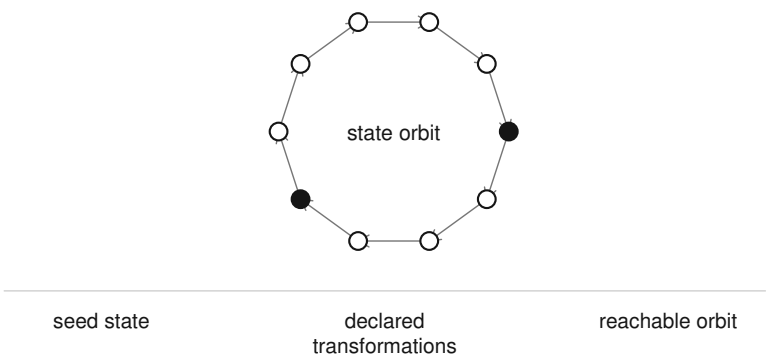
Shift-shell and suffix Hamming hunts



Finding: Search metadata-distance shells, suffix flips and literal TRAILING gaps around selected anchors.

M114 / SCHEMATIC Implemented

Synthesis transformation orbit engine



Finding: Hybrid orbit, capture/decay/catalyst/moderator transformations rank unused-byte novelty; terminology does not imply quantum physics.

M115 / SCHEMATIC Executed record

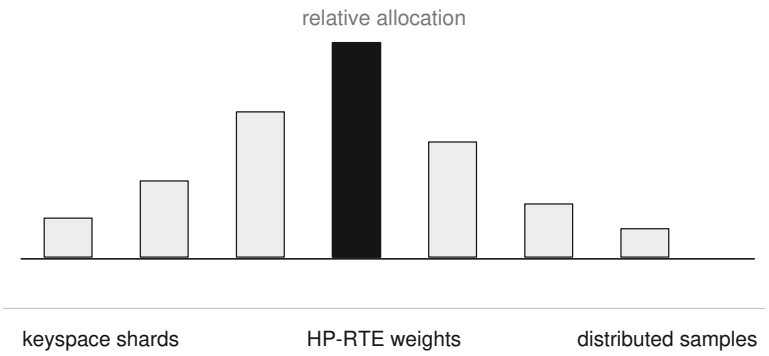
Unused-byte novelty scoring



Finding: Score 8/9 or 9/9 measures byte novelty, not target agreement. Archived 1,800-candidate verification had zero hits.

M116 / SCHEMATIC Audited/corrected

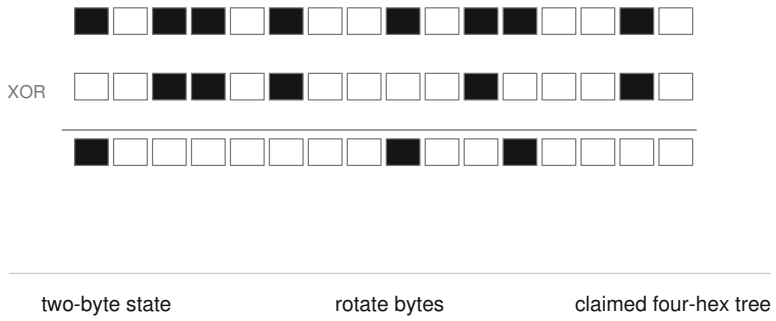
Original HP-RTE weighted shard sampling



Finding: Historical float-based biased generator excludes part of the advertised interval; corrected pipeline uses full-support integer partitions.

M117 / SCHEMATIC Audited/corrected

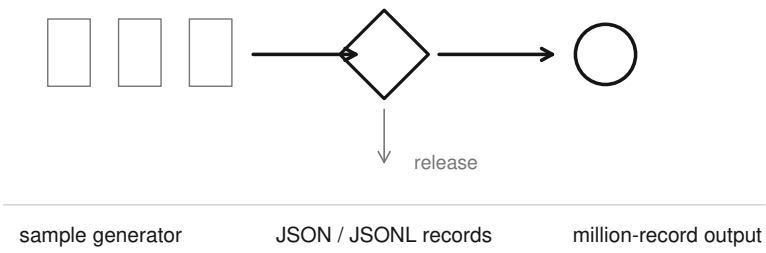
Two-byte rotation and four-hex-tree claims



Finding: Rotation exists, but original tree-depth configuration did not implement the claimed tree search.

M118 / SCHEMATIC Executed record

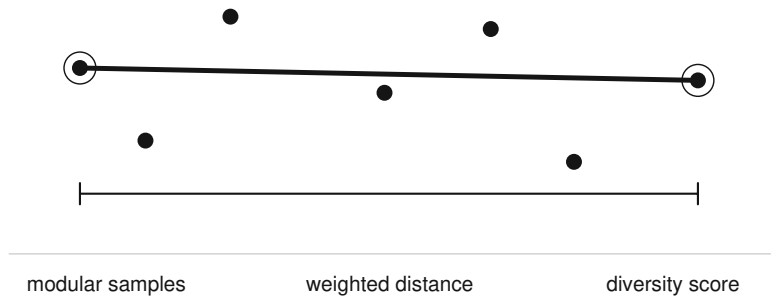
Million-record JSON/JSONL generation



Finding: Large record counts are data-generation work, not cryptographic search coverage; streaming JSONL avoids all-record memory use.

M119 / SCHEMATIC Audited/corrected

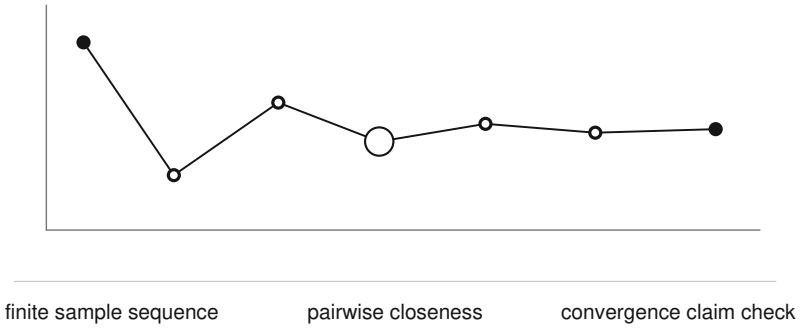
Weighted modular-diameter diversity



Finding: uint64 overflow, unsigned differences and unproved subset weights invalidate original diversity claims.

M120 / SCHEMATIC Audited/corrected

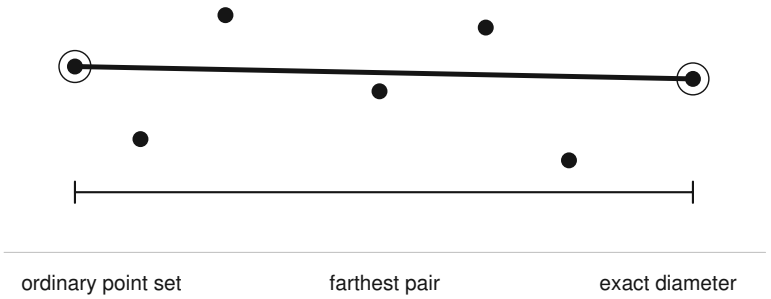
Finite-sample Cauchy/convergence claims



Finding: Sampling a finite history is not an asymptotic tail criterion; clustering cannot identify the unknown scalar.

M121 / SCHEMATIC Executed record

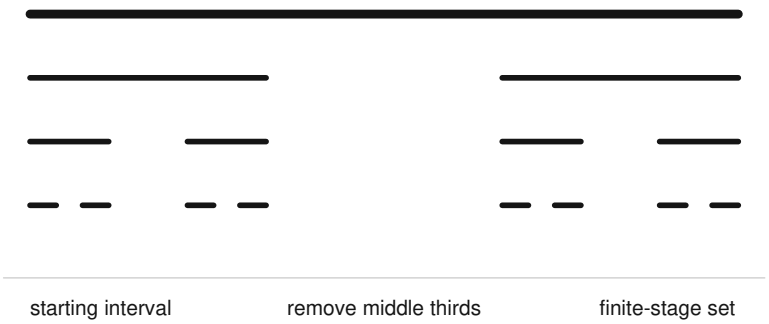
Exact ordinary diameter diagnostic



Finding: Integer max-minus-min measures sample spread without claiming convergence or cryptographic proximity.

M122 / SCHEMATIC Audited/corrected

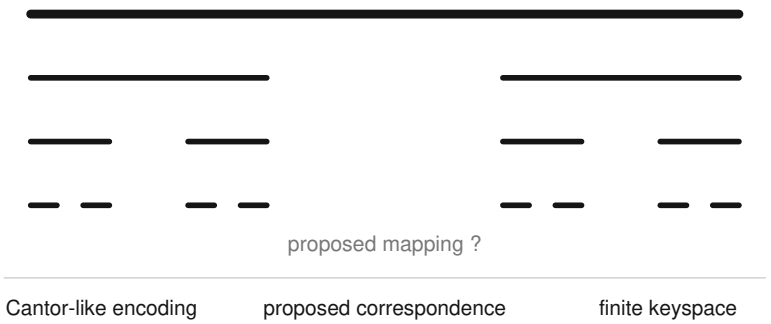
Cantor-like finite-stage constructions



Finding: Exact rational construction corrects historical measure calculations; constant half-removal tends to zero measure.

M123 / SCHEMATIC Audited/corrected

Cantor-to-keyspace homeomorphism claim



Finding: Rounding real samples to finite keys is not a homeomorphism from an infinite Cantor set.

M124 / SCHEMATIC Executed record

Byte-value arithmetic-progression avoidance



Finding: Corrected diagnostic detects nonconstant APs among byte values; it is not used to discard search candidates.

M125 / SCHEMATIC Executed record

Monochromatic byte-position progression checks



Finding: Equal bytes at equally spaced positions are distinguished from APs in the values themselves.

M126 / SCHEMATIC Audited/corrected

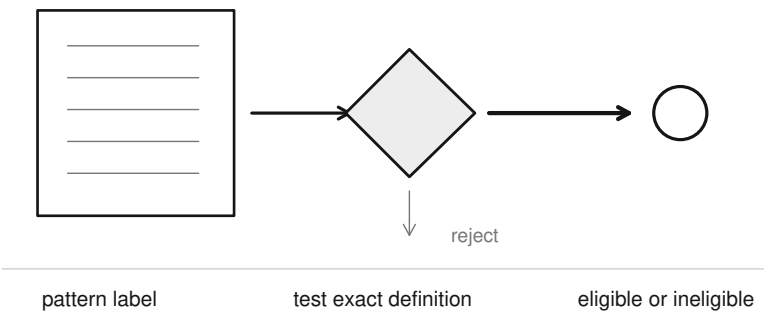
Double-AP and RamseyScript subset claims



Finding: Historical code uses placeholders and incomplete language semantics; original filter claims do not establish a cryptographic advantage.

M127 / SCHEMATIC Executed record

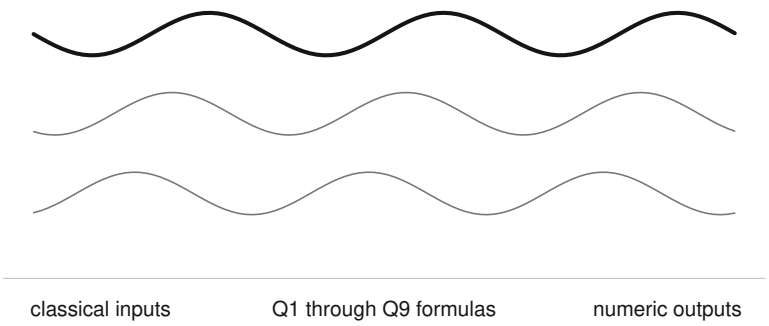
Pattern-label eligibility correction



Finding: Backtests retain all candidates while reporting pattern labels; eligibility is not evidence that a pattern predicts keys.

M128 / SCHEMATIC Audited/corrected

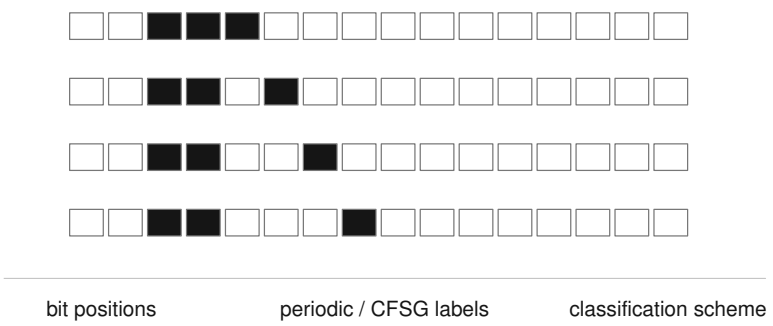
Classical hyper-phase Q1-Q9 formulas



Finding: Numerical toy formulas have no demonstrated target relation; no quantum circuit or recovery is implemented.

M129 / SCHEMATIC Conversation proposal

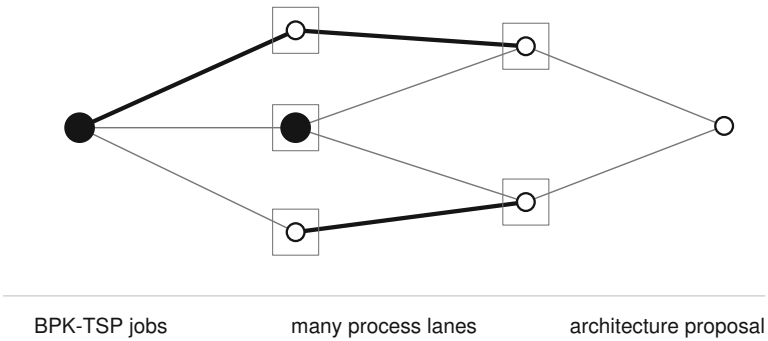
Bit-position periodic table / CFSG labels



Finding: Supplied transcript uses simulated random biases and organizational labels; no local validated cryptanalytic result was established.

M130 / SCHEMATIC Conversation proposal

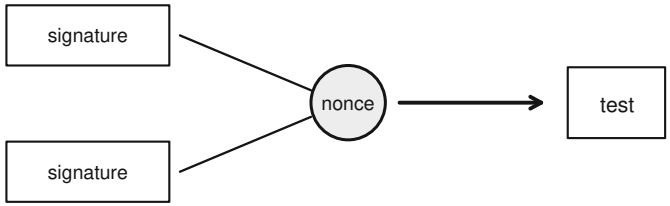
BPK-TSP and million-process architecture



Finding: Scheduling and FPGA/massive-process claims in supplied reports are proposals without corroborated execution evidence in this inventory.

M131 / SCHEMATIC Executed record

Aranyn Block-1 manual-nonce binding

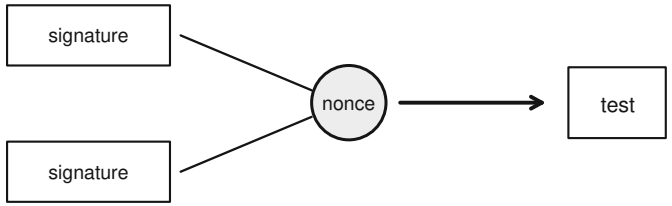


Block-1 signature | manual nonce hypothesis | binding equation check

Finding: One signature gives an affine relation between two scalar names but leaves an ECDLP-equivalent unknown.

M132 / SCHEMATIC Implemented

Puzzle-160 nonce/signing-key relation

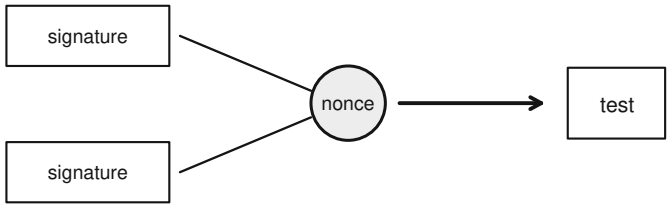


puzzle-160 signature | nonce / signing-key relation | public-point checks

Finding: Recover one scalar if the other is supplied; the published relation does not independently reveal either unknown key.

M133 / SCHEMATIC Implemented

Cross-signature #160 / Block-1 linear relations

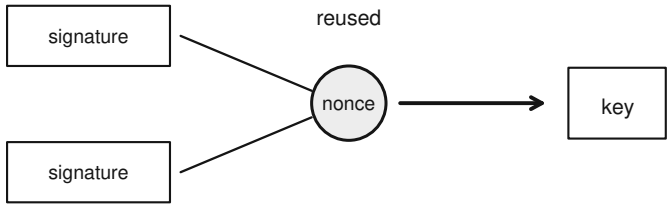


two artifact signatures | proposed linear relation | compatibility test

Finding: Search bounded small coefficients visible on public points; any result would need independent target verification.

M134 / SCHEMATIC Synthetic/recorded tests

ECDSA nonce-reuse recovery

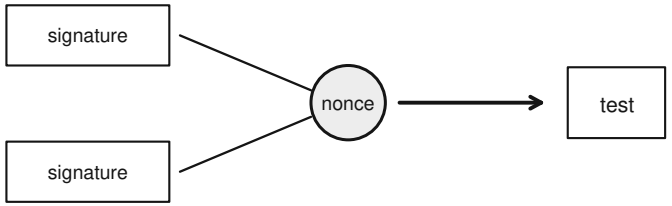


two signatures | same nonce / matching r | algebraic key recovery

Finding: Algebraic recovery works in planted examples; harvested real signatures showed no matching reuse groups in recorded samples.

M135 / SCHEMATIC Synthetic tests

Known affine-related nonce recovery

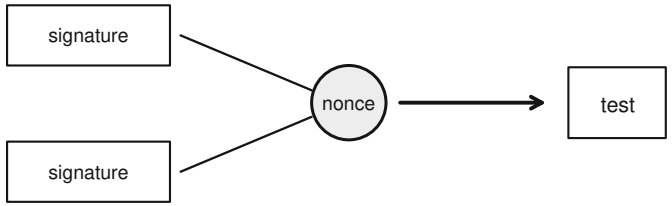


nonce pair | affine relation | linear-system recovery

Finding: Synthetic signature pairs validate the formula when the nonce relation is actually known. Applicability requires such a pair.

M136 / SCHEMATIC Executed record

Counter nonces and small offset relation tests

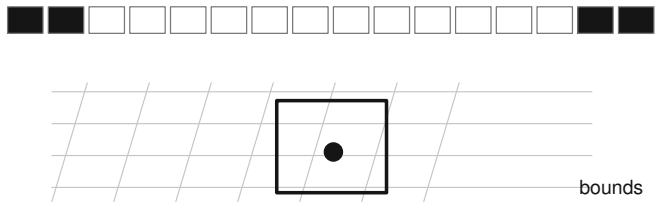


counter nonces | small offset choices | relation checks

Finding: Archived harvested-signature scans tested tiny nonces and bounded offsets; no verified wallet-key recovery reported.

M137 / SCHEMATIC Synthetic/applicability

Hidden-number/partial-MSB nonce leakage models

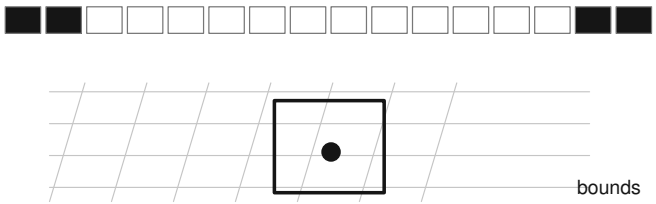


partial nonce bits | bounded error lattice | candidate validation

Finding: Synthetic bounds and applicability checks exist; the target signatures do not supply the required leakage measurements.

M138 / SCHEMATIC Applicability review

Polynonce/half-half/side-channel possibilities

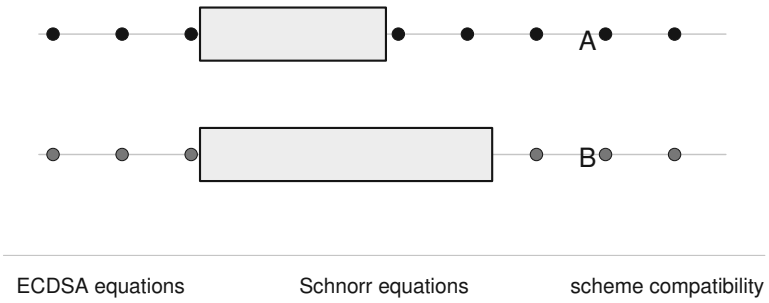


structured nonce model | additional leakage needed | recovery hypothesis

Finding: Reviewed as conditional attacks; missing signatures, special nonce structure or measurements block their use on the cited target.

M139 / SCHEMATIC Executed record

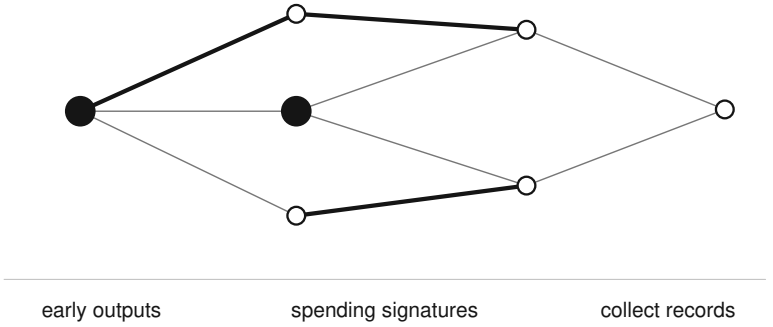
ECDSA versus Schnorr/BIP340 checks



Finding: Forced reinterpretations and challenge consistency are compared; changing signature labels does not create a new independent equation.

M140 / SCHEMATIC Executed record

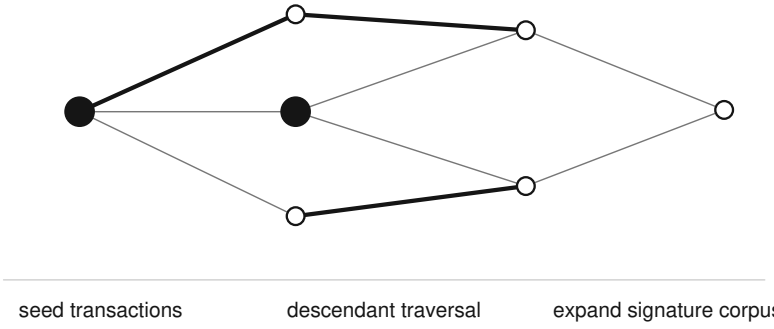
Early coinbase signature harvesting



Finding: Saved report records 51 signatures and no nonce-reuse recovery; the graph is a data source, not a solve.

M141 / SCHEMATIC Implemented

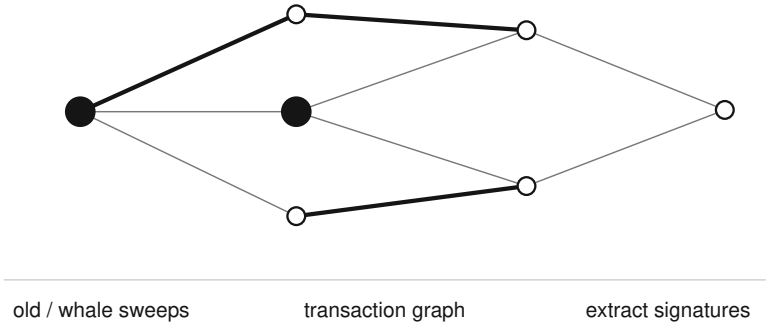
Descendant/graph signature expansion



Finding: Expand UTXO descendants and consolidation transactions to obtain additional signature samples.

M142 / SCHEMATIC Executed record

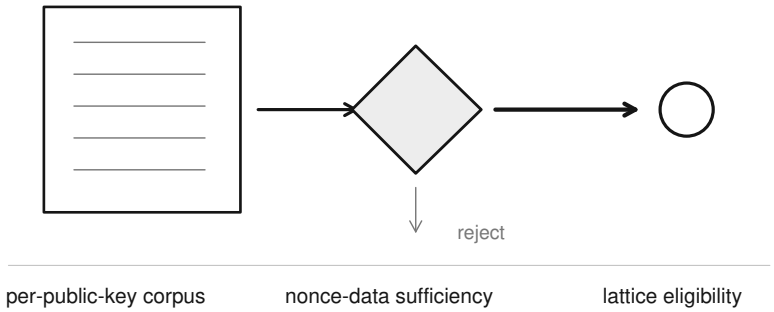
BTCparser sleeping/whale sweep signatures



Finding: Saved sweep harvested 151 signatures with no reuse groups or target-r hits in that sample.

M143 / SCHEMATIC Executed record

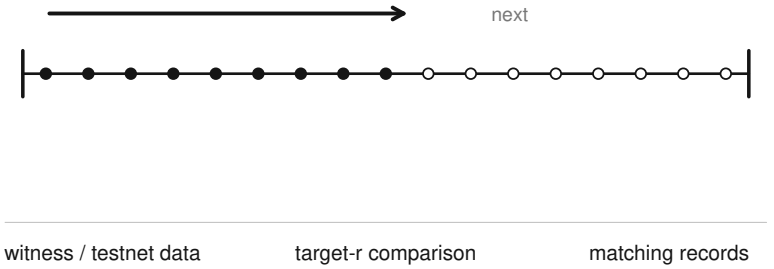
Lattice-candidate sufficiency scan



Finding: Saved report has 184 signatures but no wallet with the required multi-signature count; no lattice recovery executed.

M144 / SCHEMATIC Executed record

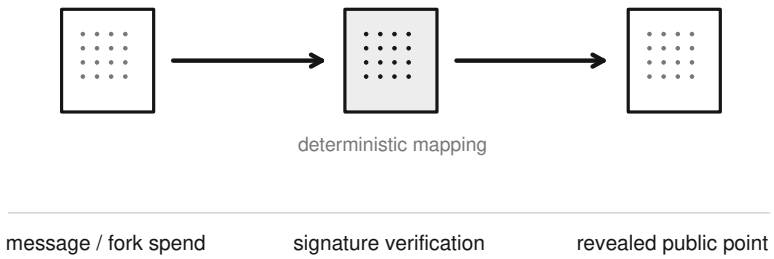
SegWit/testnet target-r scans



Finding: Search archived alternate-chain and witness signatures for the manual-nonce x-coordinate; finite scan scope only.

M145 / SCHEMATIC Implemented

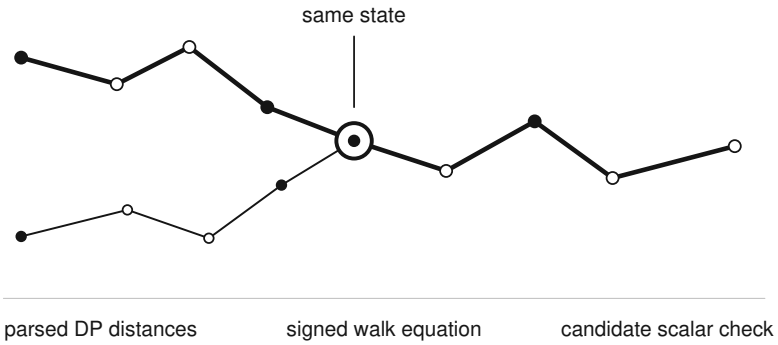
Signed-message and fork public-key leaks



Finding: Search additional public signature/pubkey material; utility depends on finding independently applicable data.

M146 / SCHEMATIC Executed record

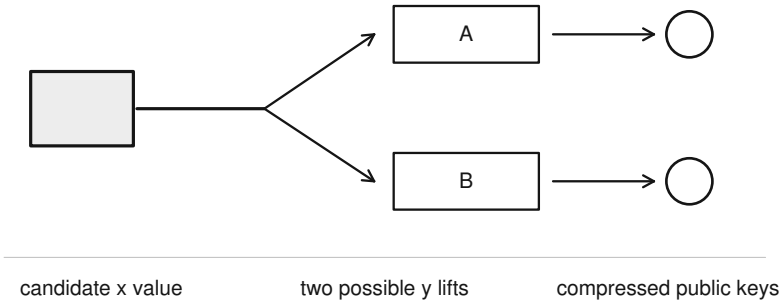
DP metadata distances and signed collision variants



Finding: Archived metadata-width, start-point, sign and collision-formula tests found no verified puzzle scalar.

M147 / SCHEMATIC Executed record

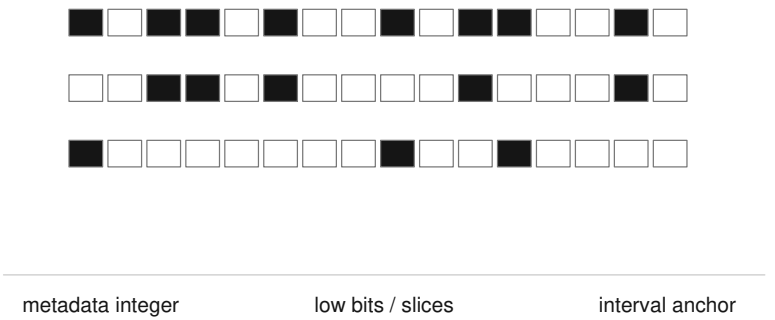
DP x-coordinates as candidate public keys



Finding: Both point parities were checked against puzzle addresses; no match in the recorded binary sample.

M148 / SCHEMATIC Executed record

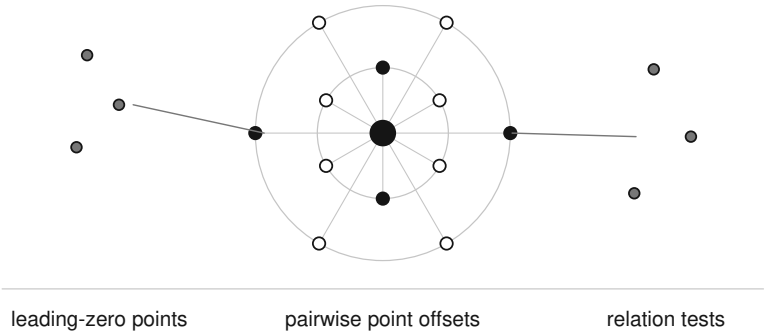
TRAILING / LOW71 / bit-slice anchors



Finding: Numerical range coincidence is separate from address correctness; recorded anchor checks failed target verification.

M149 / SCHEMATIC Implemented

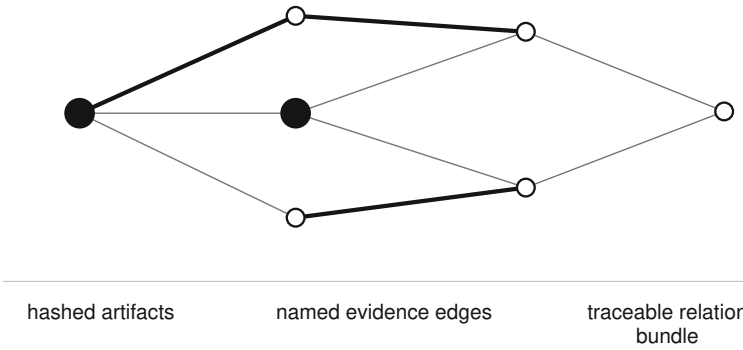
Leading-zero public-key collection analysis



Finding: Tests mask statistics, membership, pairwise differences, GLV relations, linear combinations and bounded BSGS offsets.

M150 / SCHEMATIC Implemented

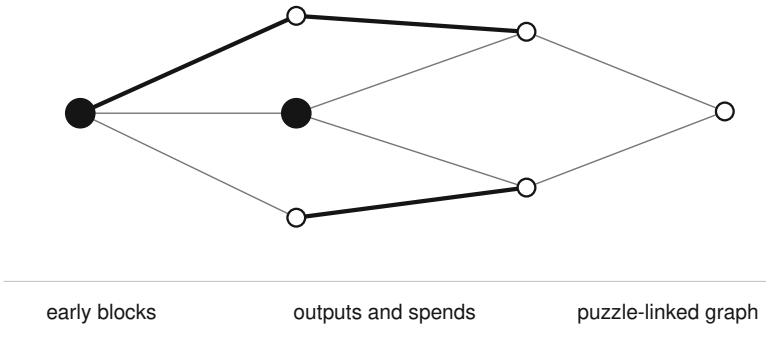
Cross-artifact relation bundles



Finding: Package DP, metadata and CTheory evidence without claiming that correlated artifacts reveal a key.

M151 / SCHEMATIC Executed record

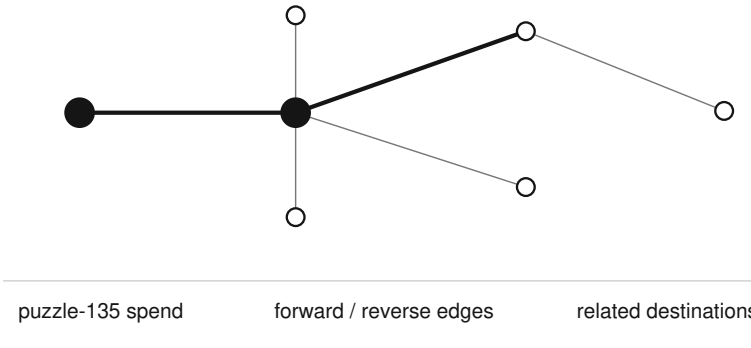
Genesis/early-coinbase relationship graph



Finding: Map published public points and spending relationships; shared transactions need not imply shared private keys.

M152 / SCHEMATIC Executed record

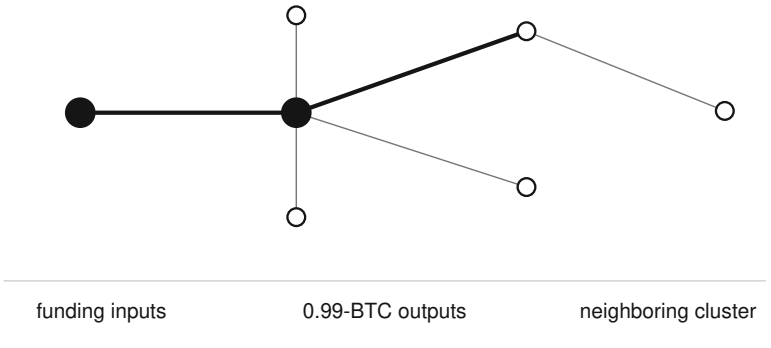
Puzzle-135 sweep/fanout tracing



Finding: Follow sweep destinations, funding and tagged puzzle neighbors; movement of funds is not local proof of solving the puzzle.

M153 / SCHEMATIC Executed record

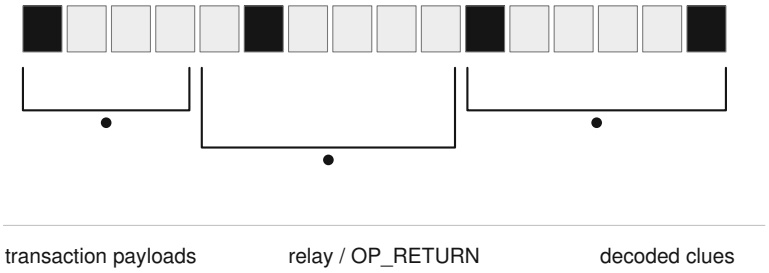
0.99-BTC fanout cluster tracing



Finding: Breadth-first funding/output graph searches and message scans are archived.

M154 / SCHEMATIC Executed record

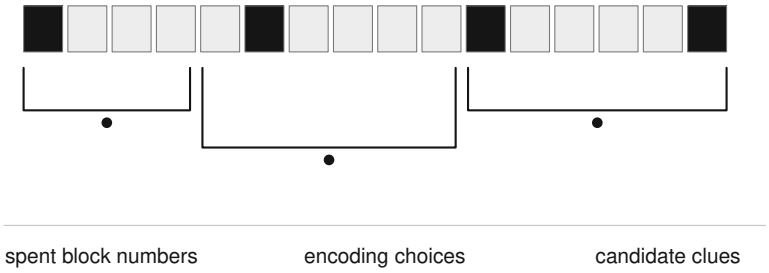
Sender relay / OP_RETURN / extra-data clues



Finding: Decode message relays and scan transaction fields for key fragments or target relations.

M155 / SCHEMATIC Implemented

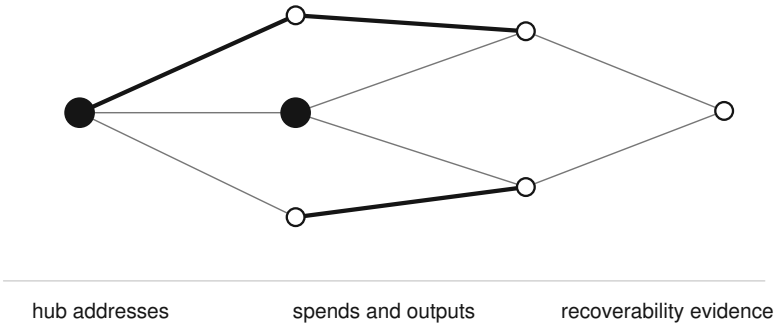
Early spent-block number encodings



Finding: Test block-index encodings and inspect spends for puzzle-like hexadecimal data.

M156 / SCHEMATIC Executed record

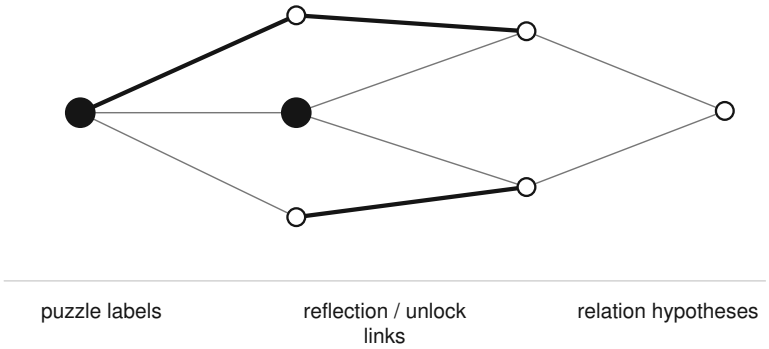
Hub-address and recoverability scans



Finding: Address/UTXO activity scans identify data leads; no present ownership or balance conclusions are newly verified here.

M157 / SCHEMATIC Executed record

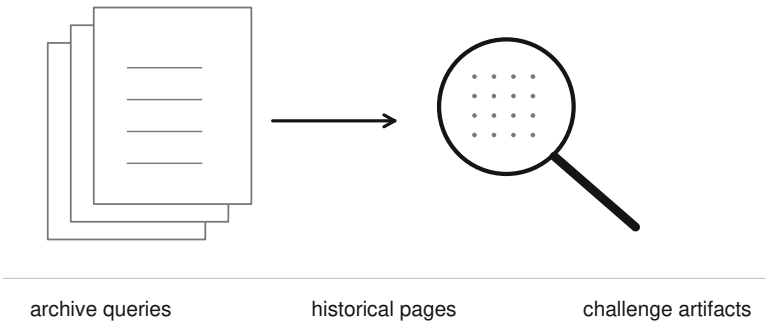
Puzzle reflection and unlock relationship scans



Finding: Compare puzzle labels, addresses and transaction relationships; no validated universal unlocking rule established.

M158 / SCHEMATIC Executed record

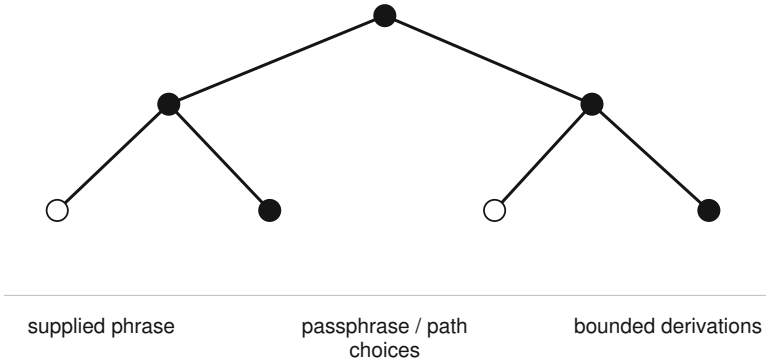
Historical website/archive and directory searches



Finding: Old Bitcoin sites, indexes and Drive leads were triaged for relevant artifacts.

M159 / SCHEMATIC Implemented/branch record

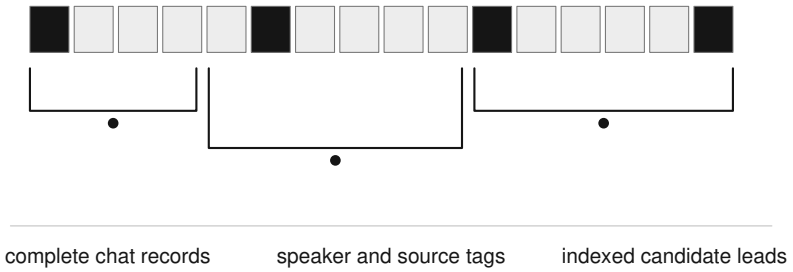
Time-Will-Tell phrase/path hypotheses



Finding: Bounded mnemonic subset, passphrase and derivation-path tests are documented; private phrases are excluded from this report.

M160 / SCHEMATIC Executed record

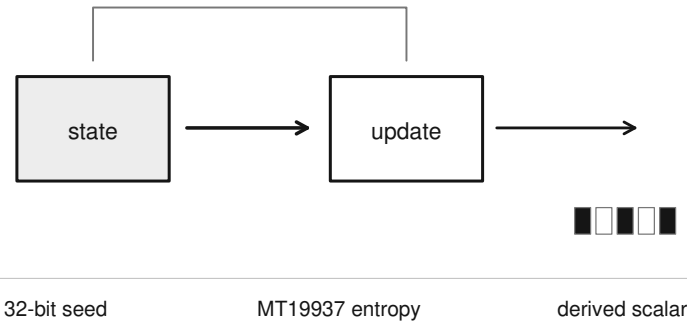
Complete chat-lead indexing and candidate extraction



Finding: Existing speaker-aware source index covers 685 matching conversations; assistant assertions are not execution evidence.

M161 / SCHEMATIC Recorded GPU sweep

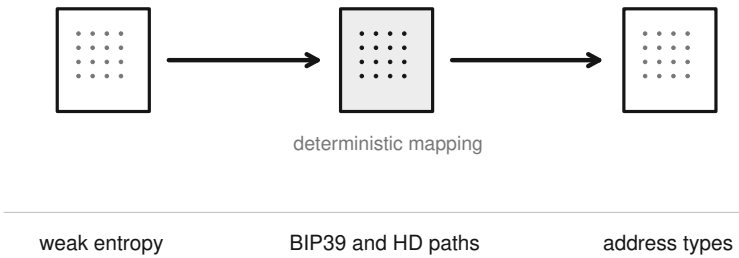
Milk Sad raw MT19937 seed expansion



Finding: Archived report claims full 32-bit seed sweep with zero funded-index hits; this inventory does not rerun or independently certify that sweep.

M162 / SCHEMATIC Recorded GPU sweep

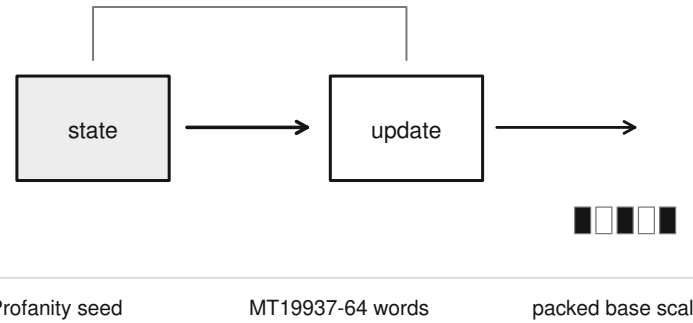
Milk Sad BIP39/BIP44/49/84 derivations



Finding: Archived mnemonic and derivation-path sweep reports zero funded-index hits; bounded historical index and path coverage only.

M163 / SCHEMATIC Recorded GPU sweep

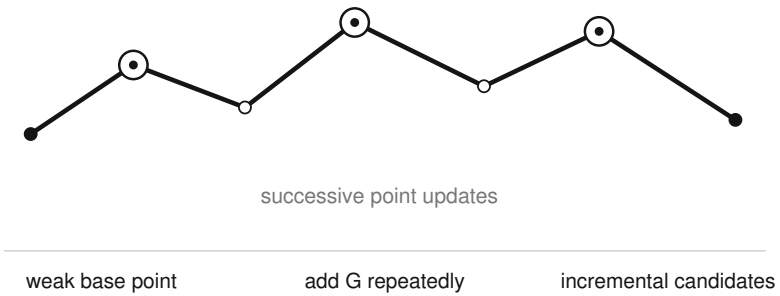
Profanity MT19937-64 base-key replay



Finding: Archived report enumerates both word orders over 32-bit seeds with zero funded-index hits at offset zero.

M164 / SCHEMATIC Branch/implementation

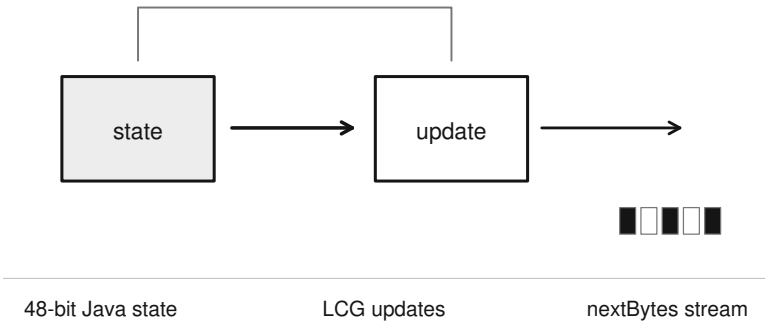
Profanity incremental offset grinding



Finding: Extends base keys with small scalar offsets. Branch notes explicitly say real-GPU compile/verification was not yet done there.

M165 / SCHEMATIC Implemented/registry

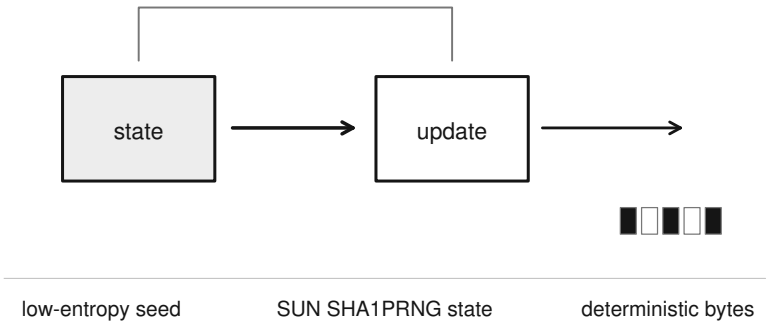
Java util.Random replay



Finding: Models the 48-bit LCG and timestamp-seeded nextBytes path; registered runtime vectors distinguish model fidelity from a recovery.

M166 / SCHEMATIC Implemented/registry

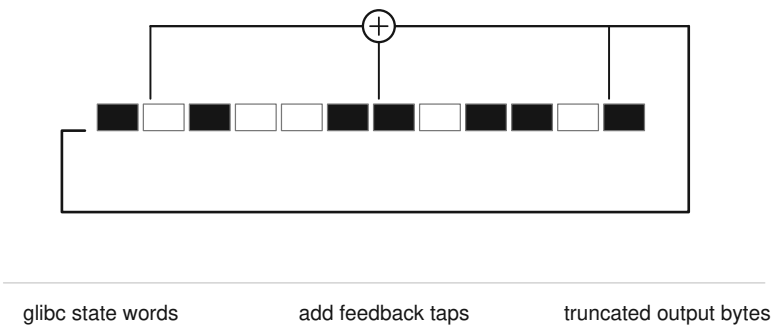
Java SUN SHA1PRNG low-entropy seed replay



Finding: Deterministic setSeed model is present; not every Java/Android wallet follows that path.

M167 / SCHEMATIC Implemented/registry

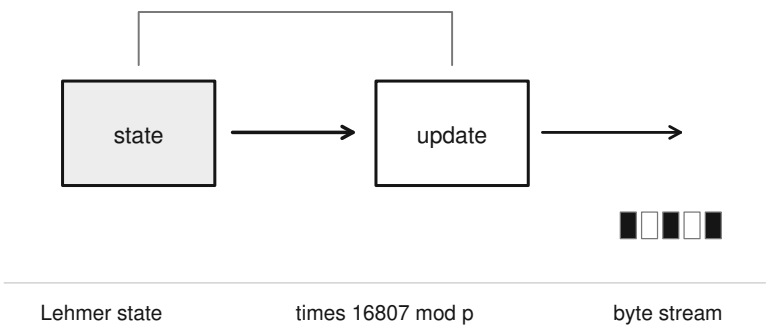
glibc rand/random byte-fill replay



Finding: Models clock-seeded additive-feedback output truncated into key bytes.

M168 / SCHEMATIC Implemented/registry

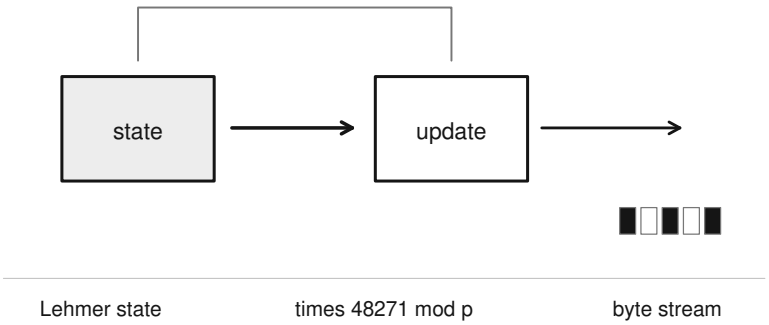
Lehmer minstd_rand0 replay



Finding: Models multiplier 16807 weak-generator key bytes.

M169 / SCHEMATIC Implemented/registry

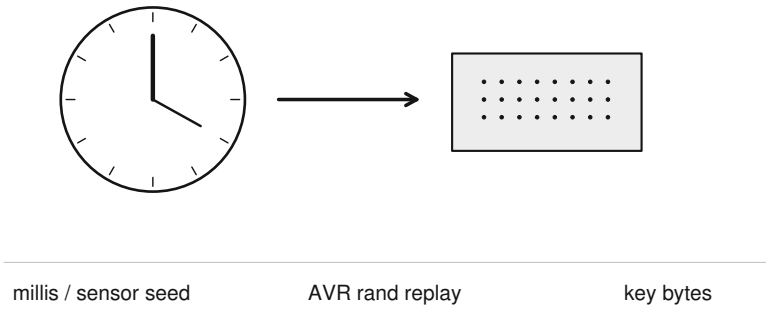
Lehmer minstd_rand replay



Finding: Models multiplier 48271 key-byte generation, separately from minstd_rand0.

M170 / SCHEMATIC Implemented/registry

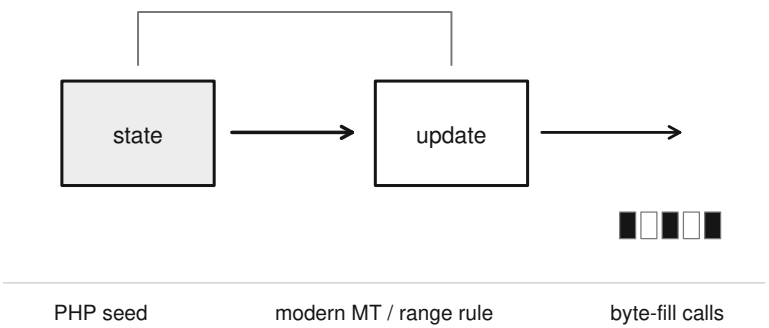
AVR/Arduino rand replay



Finding: Models weak embedded Lehmer-style output under enumerable seeds.

M171 / SCHEMATIC Implemented/registry

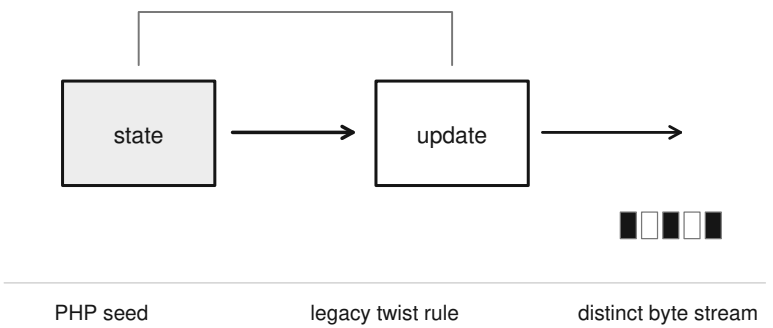
Modern PHP MT19937 byte-fill replay



Finding: PHP 7.1+ generator/range semantics are modeled separately from legacy PHP.

M172 / SCHEMATIC Implemented/registry

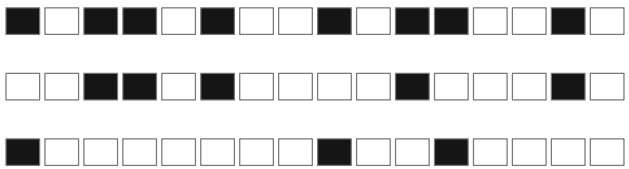
Legacy PHP broken-twist replay



Finding: Models the older mt_rand twist and range-scaling behavior, not covered by ordinary MT19937 assumptions.

M173 / SCHEMATIC Implemented/registry

PHPCoinAddress legacy and modern keygen

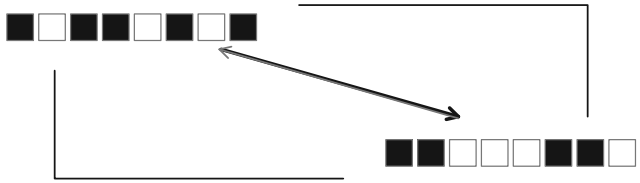


first-byte constraint | wallet-specific calls | key byte string

Finding: Dedicated first-byte/range rules are modeled under both PHP generator modes.

M174 / SCHEMATIC Implemented/registry

V8 MWC1616 / Randstorm-era byte fill

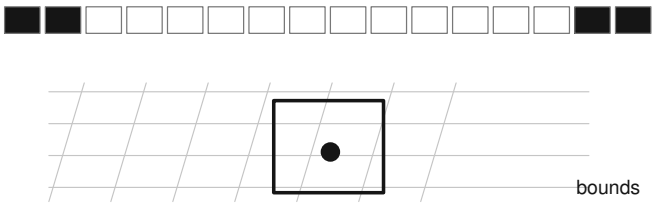


two V8 states | multiply-with-carry | combined outputs

Finding: Historical JavaScript Math.random byte generation model; a vulnerable-family model is not attribution to a target wallet.

M175 / SCHEMATIC Implemented

V8 xorshift128 state recovery

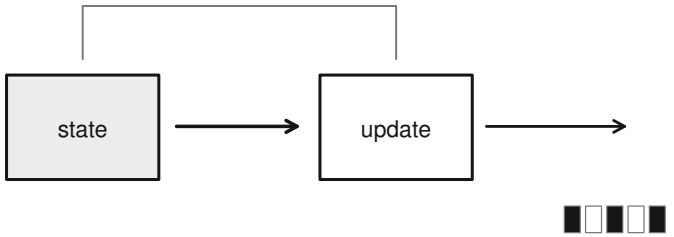


observed random outputs | xorshift constraints | compatible states

Finding: Recovers generator state from observed Math.random outputs under the modeled engine semantics; no target observations established here.

M176 / SCHEMATIC Implemented/registry

.NET System.Random replay

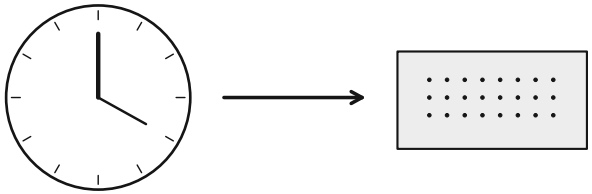


TickCount-style seed | .NET subtractive state | replayed outputs

Finding: Models explicitly seeded subtractive generator and TickCount-style seed scope.

M177 / SCHEMATIC Implemented

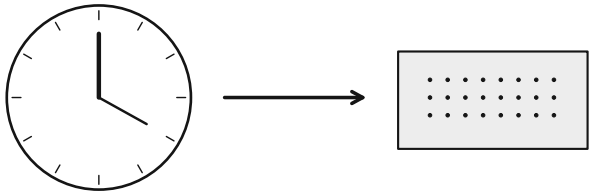
Timestamp/UUID hash-derived keys



Finding: SHA-256 clock, millisecond and weak-UUID key recipes are modeled as bounded generator hypotheses.

M178 / SCHEMATIC Executed record

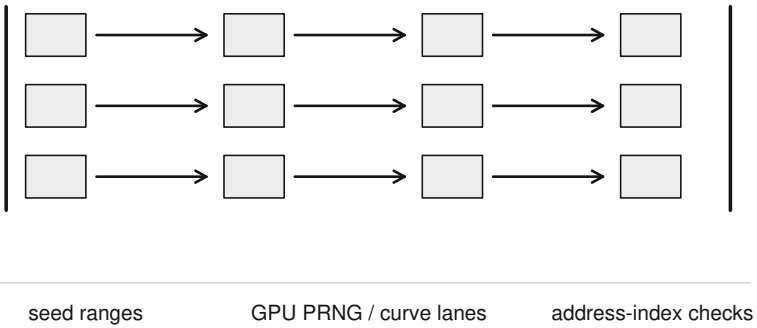
Early-2009 clock-window seed replay



Finding: Widest saved report records 250,581,618 tested keys and no early-coinbase target recovery; true machine-state entropy is not enumerated.

M179 / SCHEMATIC Recorded GPU sweep

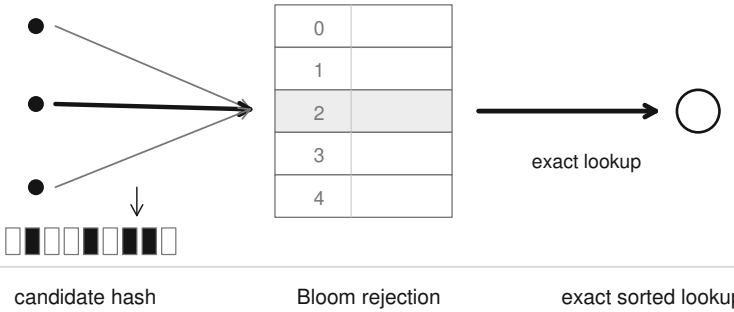
Weak-entropy multi-family CUDA sweep



Finding: Saved report claims 27.5 billion seeds and zero funded-index hits, with planted-index detection tests. Historical record, not rerun here.

M180 / SCHEMATIC Implemented

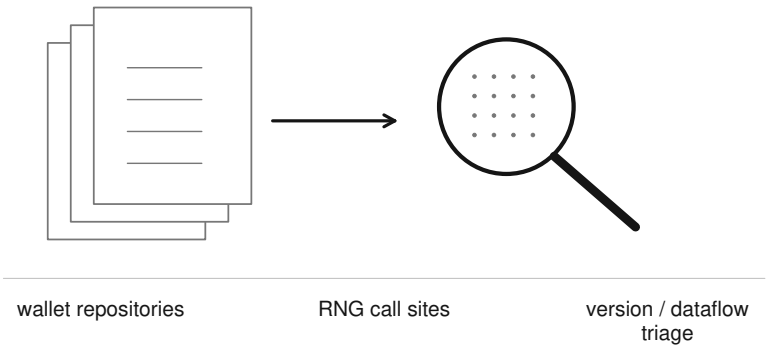
Bloom plus sorted funded-address index



Finding: Probabilistic prefilter backed by exact lookup; index date and supported address types bound conclusions.

M181 / SCHEMATIC Executed record

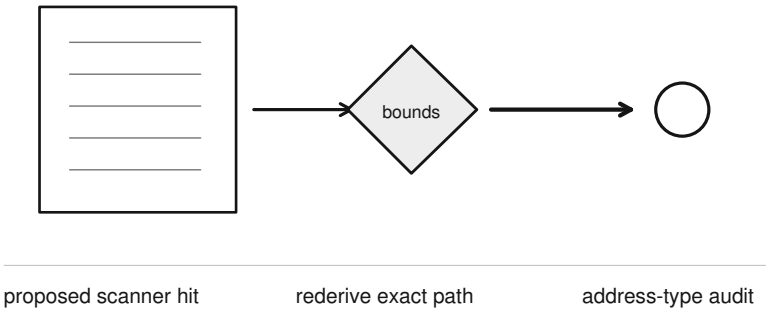
Weak-RNG repository-source triage



Finding: Pattern hits and disclosures locate candidate implementations; historical/current refs and dependency bugs must be distinguished.

M182 / SCHEMATIC Audited/corrected

Recoverable-hit rederivation/address-type audit



Finding: Recompute candidate addresses and reject type mismatches before reporting a recovery; prior raw hits were not all valid.

M183 / SCHEMATIC Executed record

Digital-root puzzle-index association



Finding: One of thirteen archived creative theories; report marks it nonsignificant, not a recovered-key method.

M184 / SCHEMATIC Executed record

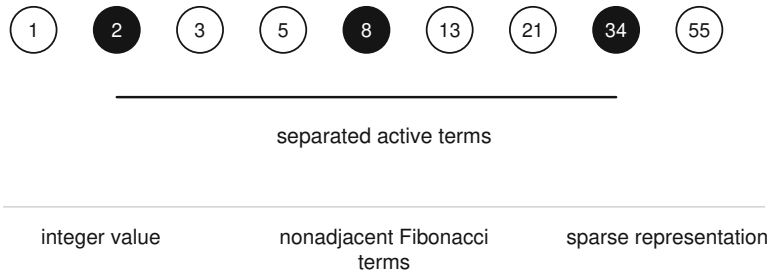
Public-key x-coordinate steganography



Finding: Compare key/public-coordinate low-bit relations with random controls; no convincing result recorded.

M185 / SCHEMATIC Executed record

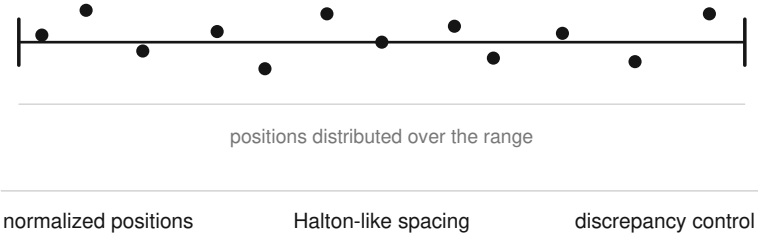
Zeckendorf/Fibonacci sparsity



Finding: Compare Fibonacci representation weights with random keys; no significant anomaly reported.

M186 / SCHEMATIC Executed record

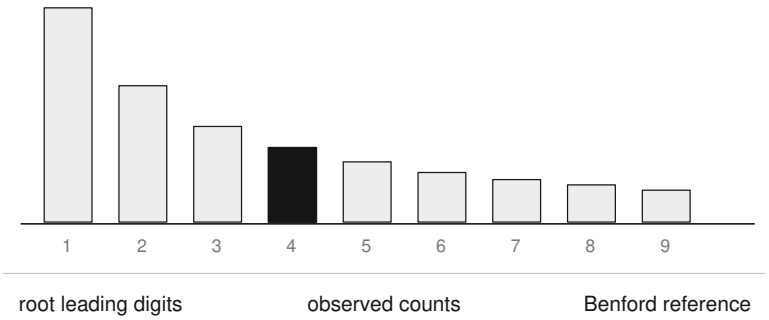
Low-discrepancy / Halton-like tier placement



Finding: Compare normalized-position discrepancy; no significant deterministic schedule established.

M187 / SCHEMATIC Executed record

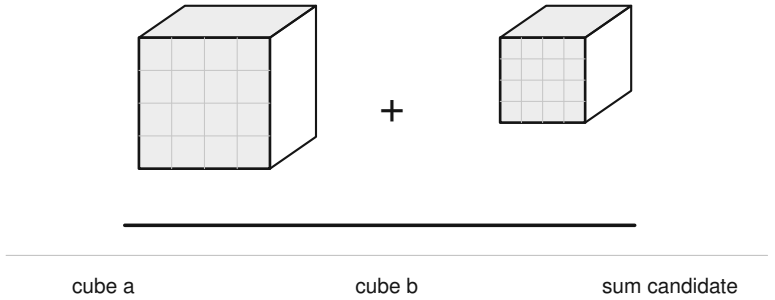
Benford leading digits of roots



Finding: Root first-digit distribution compared with controls; no convincing signal reported.

M188 / SCHEMATIC Executed record

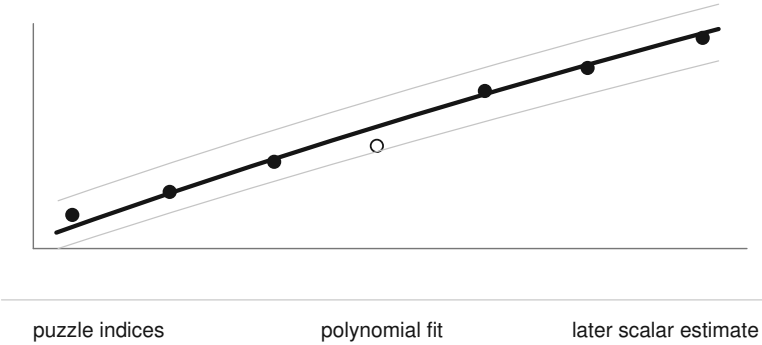
Taxicab sums of two cubes



Finding: Search special cube-sum representations of known keys; no transferable high-index key relation.

M189 / SCHEMATIC Executed record

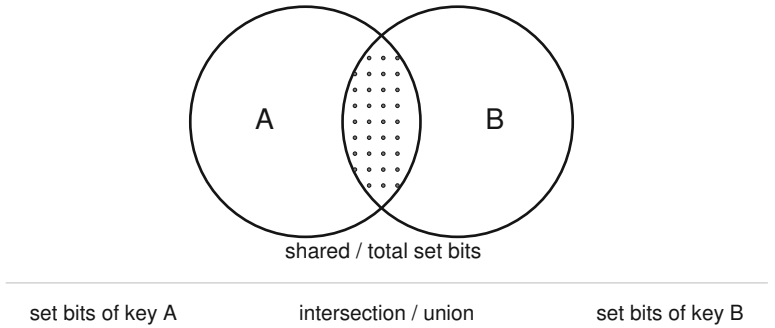
Polynomial in puzzle number



Finding: Leave-one-out prediction compared with random keys; fitting known data does not determine the next scalar.

M190 / SCHEMATIC Executed record

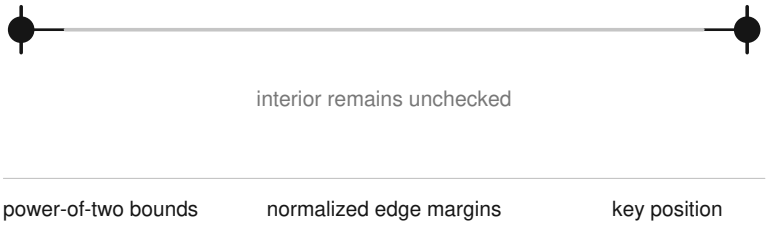
Consecutive bit-Jaccard similarity



Finding: Bit-set overlap compared with random controls; no significant inheritance relation reported.

M191 / SCHEMATIC Executed record

Mersenne-adjacent margin hypothesis



Finding: Distances to power-of-two boundaries compared with random ranges.

M192 / SCHEMATIC Executed record

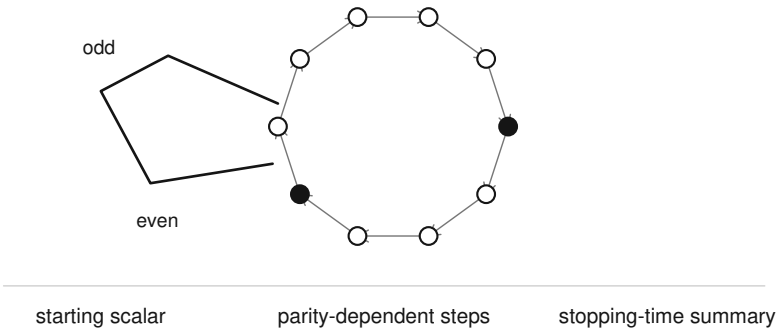
Shared CRT residue / digital modular biases



Finding: Small-modulus residue frequencies examined; exploratory bias is not proof of a target constraint.

M193 / SCHEMATIC Executed record

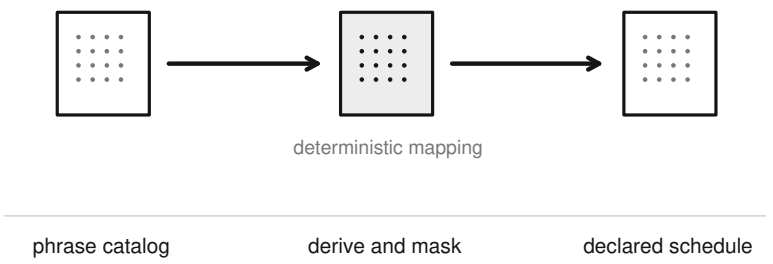
Collatz stopping-time association



Finding: Compare stopping statistics with random-key controls; no predictive result reported.

M194 / SCHEMATIC Executed record

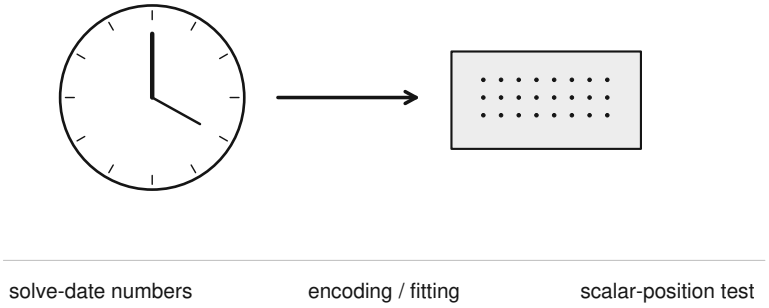
Brainwallet/BIP39 phrase schedule



Finding: Catalog phrase-derived keys and masked tiers tested; archived creative report records no nontrivial hits.

M195 / SCHEMATIC Executed record

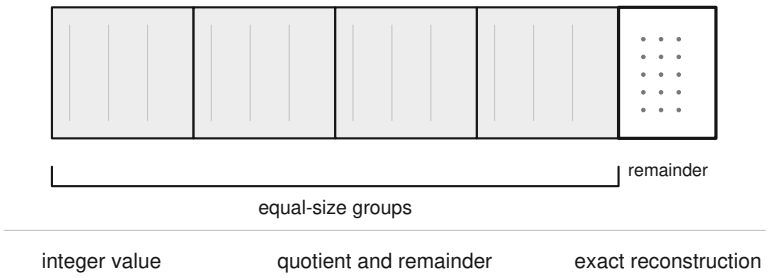
Solve-date scalar encoding



Finding: Test date patterns in keys; no significant association recorded.

M196 / SCHEMATIC Executed record

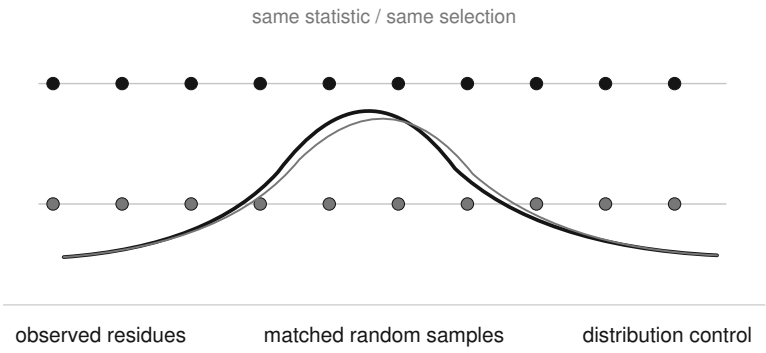
UniMath-inspired arithmetic identities



Finding: Exact division, reconstruction, congruence and remainder assertions on known keys; arithmetic checks are not an Agda proof or prediction.

M197 / SCHEMATIC Implemented

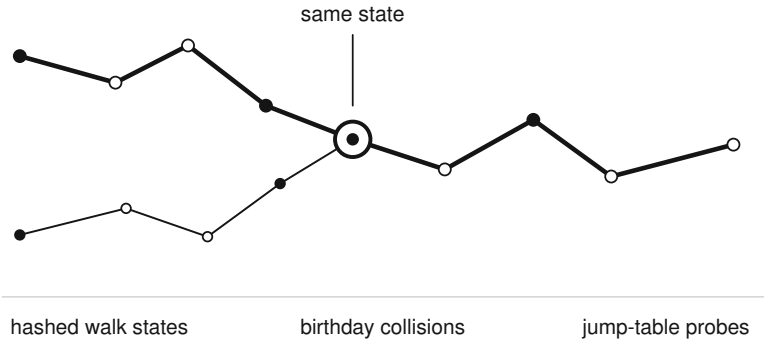
Residue/Legendre/coefficient distribution controls



Finding: Random controls and KS/chi-square-style summaries distinguish transformation artifacts from candidate key bias.

M198 / SCHEMATIC Implemented

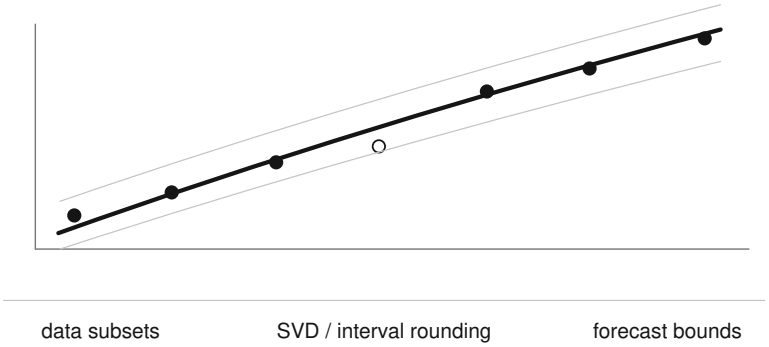
DP hashing/birthday/URAND jump probes



Finding: Bitsavers-inspired hash-table, collision and jump-distribution experiments assess solver mechanics.

M199 / SCHEMATIC Implemented

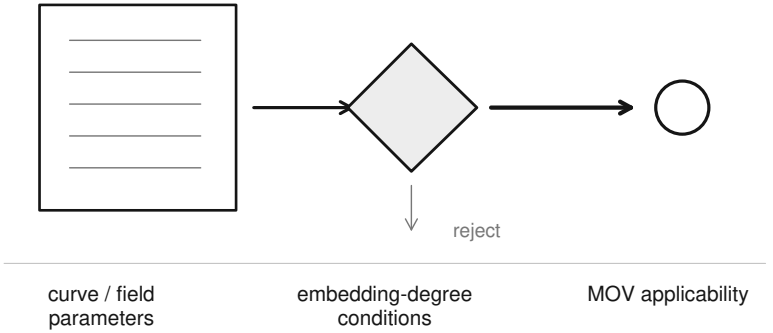
Regression SVD/subset/interval arithmetic variants



Finding: Historical numerical-method variants audit R2 fits and rounding; not new information about the target.

M200 / SCHEMATIC Applicability review

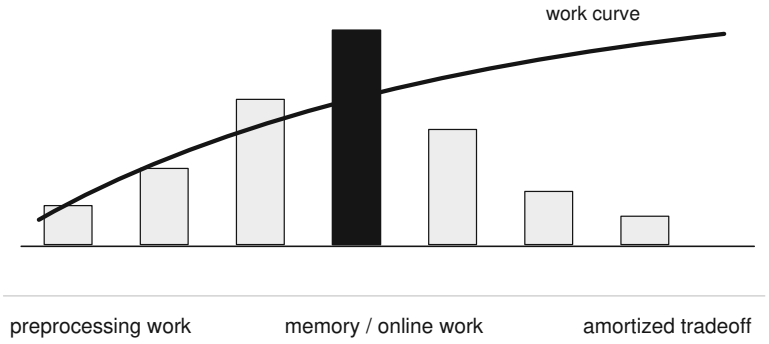
MOV and finite-field index-calculus applicability



Finding: Registry checks distinguish attacks requiring unsuitable curve structure from applicable interval methods.

M201 / SCHEMATIC Synthetic/applicability

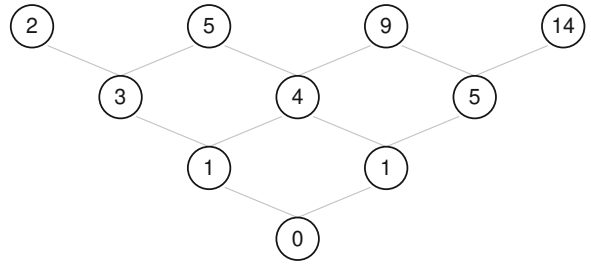
DDL preprocessing tradeoff bounds



Finding: Analysis estimates restricted discrete-log tradeoffs under additional setup assumptions; no blind puzzle recovery demonstrated.

M202 / SCHEMATIC Executed record

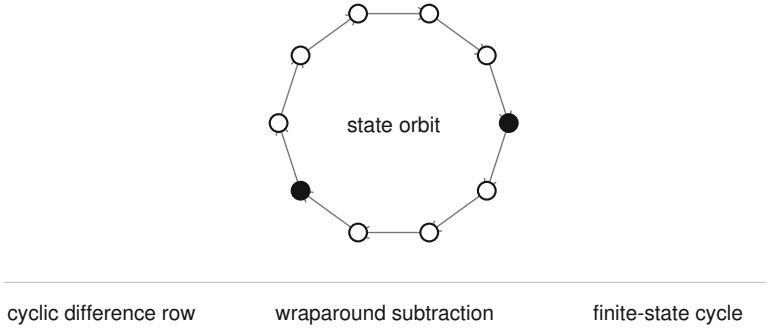
Finite modular difference triangles



Finding: Exact finite-row experiments track polynomial/exponential residues through all levels; they do not reconstruct an unknown puzzle key.

M203 / SCHEMATIC Executed record

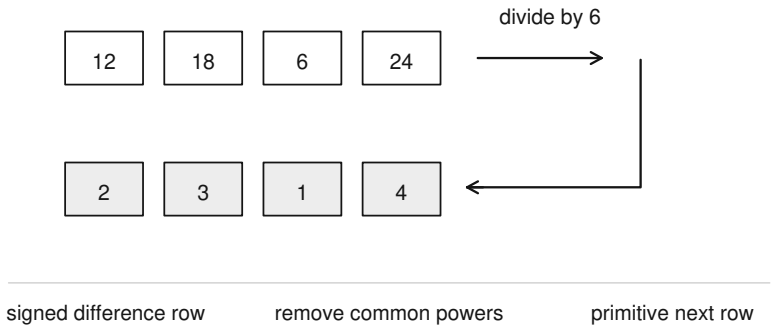
Cyclic difference orbit classification



Finding: Exhaustive finite study classified 19,617 starting rows over 35 configurations into zero/cycles.

M204 / SCHEMATIC Executed record

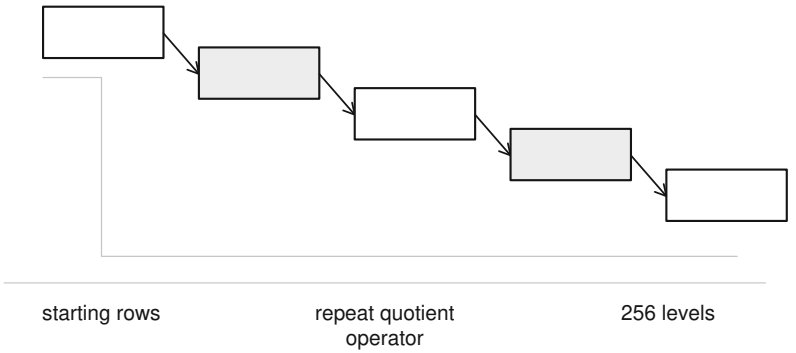
Signed quotient hierarchy with common-factor removal



Finding: Remove factors of the full modulus before continuing signed differences; zero, cycle and capped outcomes remain separate.

M205 / SCHEMATIC Executed record

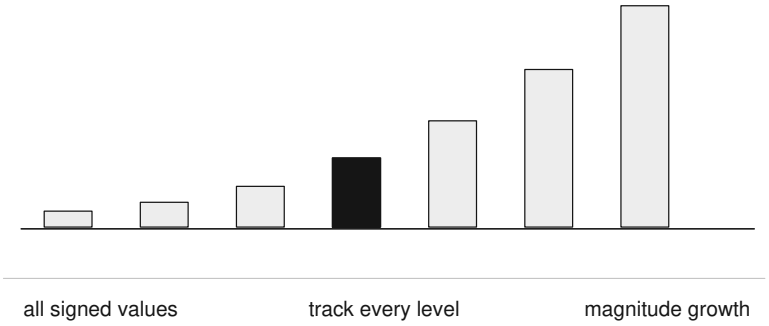
256-level quotient continuation



Finding: Saved survey records 163 zero, 600 cycles and 18,854 unresolved rows; no new resolutions after the earlier cap.

M206 / SCHEMATIC Executed record

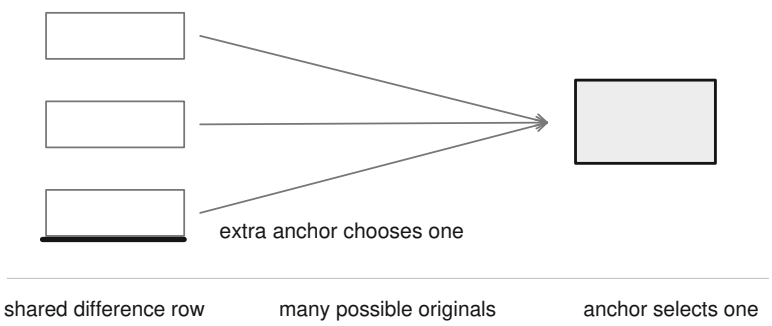
All-value magnitude/growth tracking



Finding: More than 23 million entry observations distinguish growth from termination; unresolved finite caps are not nontermination proofs.

M207 / SCHEMATIC Executed record

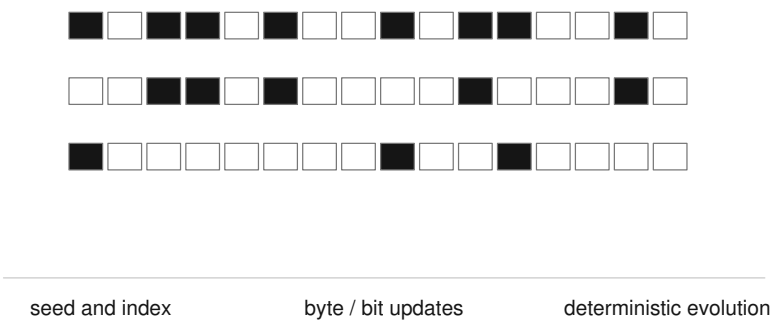
Inverse-difference kernel/anchor tests



Finding: Enumerate compatible starting rows; anchors can resolve lost constants, but the residual hierarchy supplies no missing observations.

M208 / SCHEMATIC Implemented

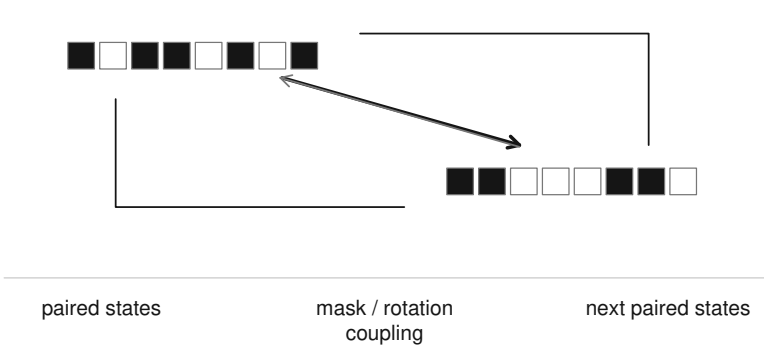
Deterministic index, byte and bit evolution



Finding: Unified framework includes index mapping, linear increments, byte/bit steps and monotonic traversal; code is not an unknown-key predictor.

M209 / SCHEMATIC Implemented

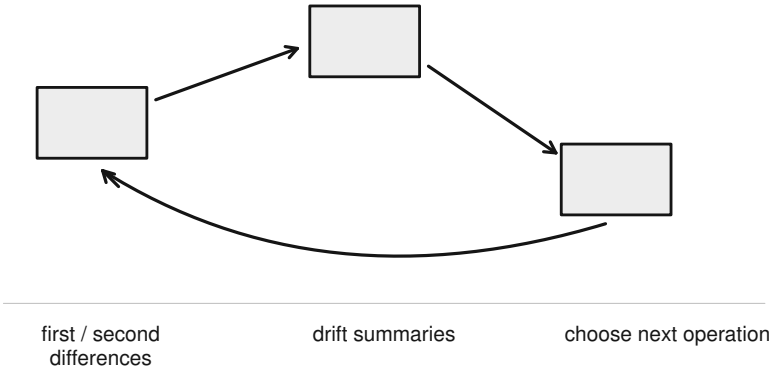
Masking, rotation and coupled sequences



Finding: Unified framework includes mask application, dual-sequence coupling, mask rotation and structured 256-step schedules.

M210 / SCHEMATIC Implemented

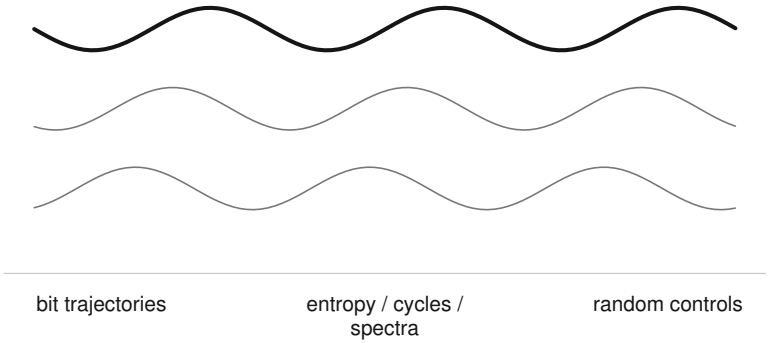
First/second-difference and drift schedulers



Finding: Analyze trajectory differences or switch transformations using entropy thresholds; diagnostic evolution is not key recovery.

M211 / SCHEMATIC Executed record

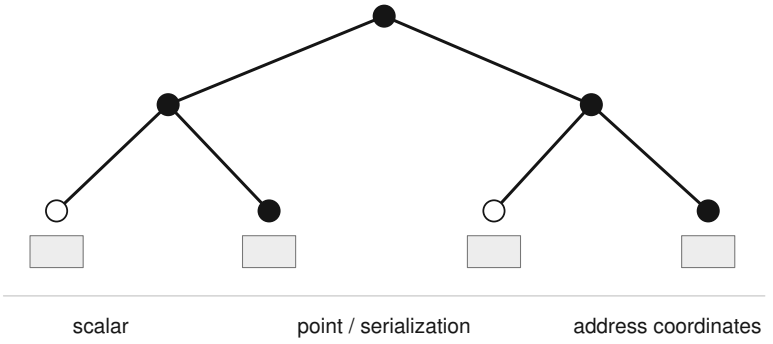
Entropy, cycles, compression and spectral diagnostics



Finding: Compare deterministic trajectories with uniform controls, including cycle tests, spacings and spectral summaries; no quantum execution implied.

M212 / SCHEMATIC Implemented

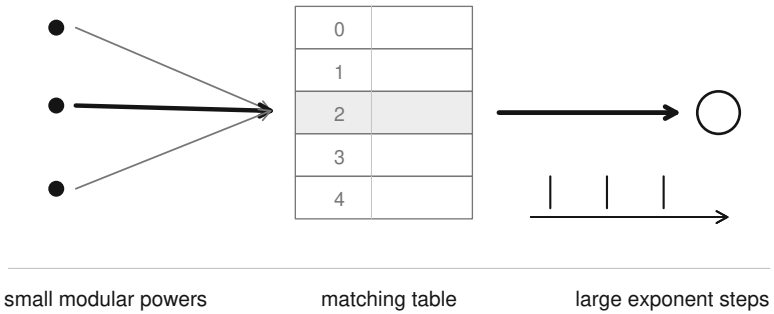
Four-dimensional key/public/address visualization



Finding: Display scalar-to-point-to-hash relationships in alternate layouts; no invertible address-to-scalar map is established.

M213 / SCHEMATIC Executed record

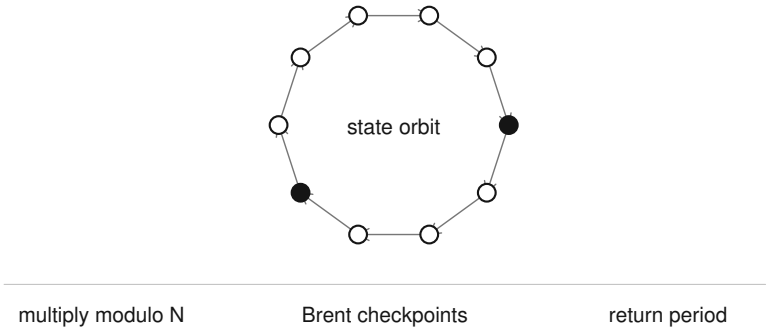
Classical BSGS multiplicative-order finder



Finding: Exact small-integer order traces and backend comparisons; stands in for an order oracle, not a quantum device.

M214 / SCHEMATIC Executed record

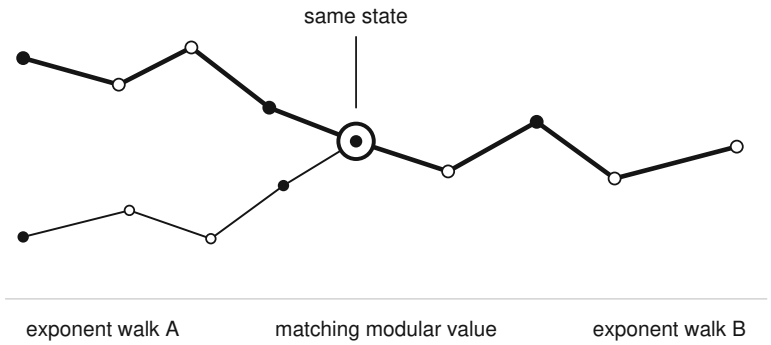
Brent cycle-detection order finder



Finding: Classical linear-period comparison backend; measured on bounded small semiprimes.

M215 / SCHEMATIC Executed record

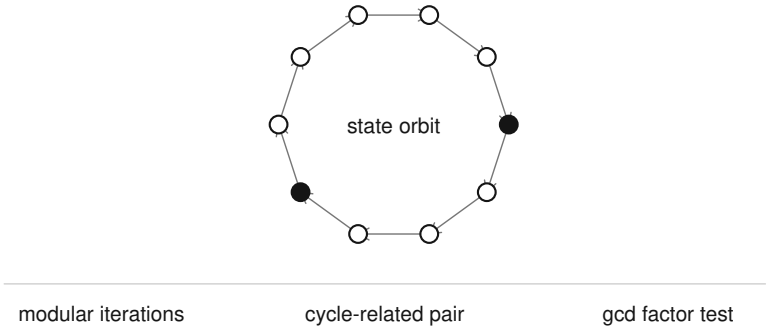
Kangaroo multiplicative-order finder



Finding: Expanding exponent-interval method compared with BSGS and Brent; not a secp256k1 scalar recovery.

M216 / SCHEMATIC Executed record

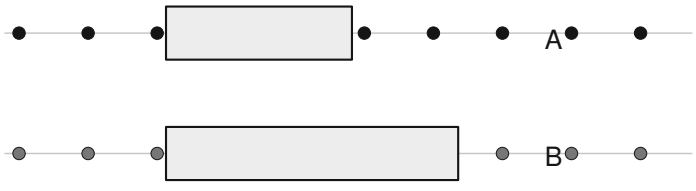
Pollard-rho factoring fast path



Finding: Canonical factor command reports its shortcut explicitly; factoring 3233 must not be relabeled as a quantum Shor run.

M217 / SCHEMATIC Executed record

Python/GMP/Rust controlled order benchmarks

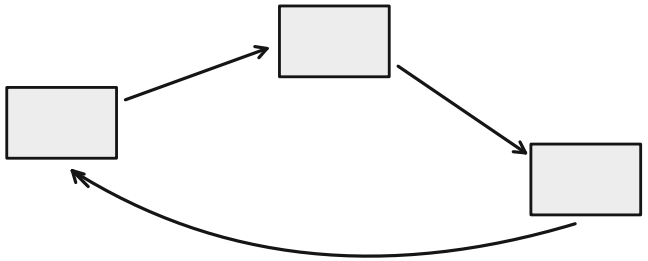


same modulus and base | Python / GMP / Rust | order and time checks

Finding: Shared modulus/base backend matrix records finite speedups; source/runtime differences are pinned in historical manifests.

M218 / SCHEMATIC Executed record

Adaptive Shor simulation trial budgets

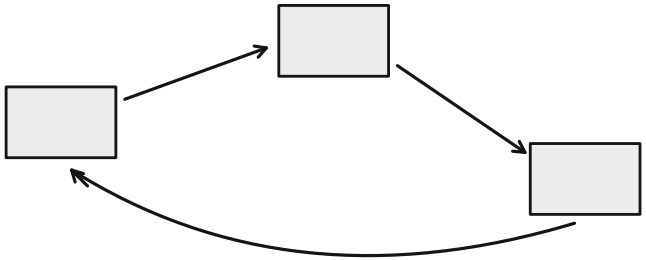


trial outcomes | estimate success rate | adapt repetition budget

Finding: Empirical/analysis/improvement loop adjusts trial counts with Wilson-bound and runtime metrics.

M219 / SCHEMATIC Branch/implementation

UCB/EWMA/early-stop method selection

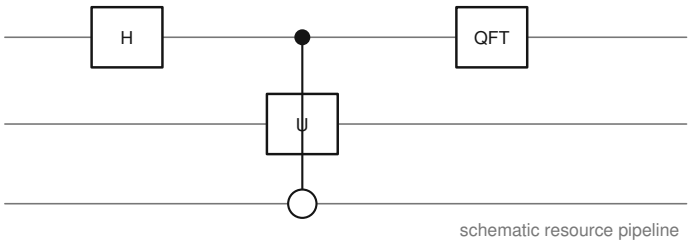


backend reward streams | UCB / EWMA policy | explore or stop

Finding: Archive describes adaptive backend policy and smoothing; simulator optimization is not target-scale quantum cryptanalysis.

M220 / SCHEMATIC Audited/corrected

Semiprime-per-puzzle circuit/manifest analogy

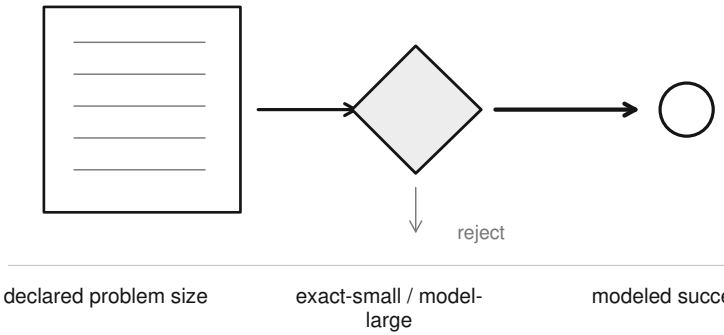


puzzle-derived semiprime | order / gcd pipeline | resource analogy

Finding: Factoring challenges derived from puzzle data are resource-scale analogies, not reductions recovering puzzle scalars.

M221 / SCHEMATIC Implemented/model

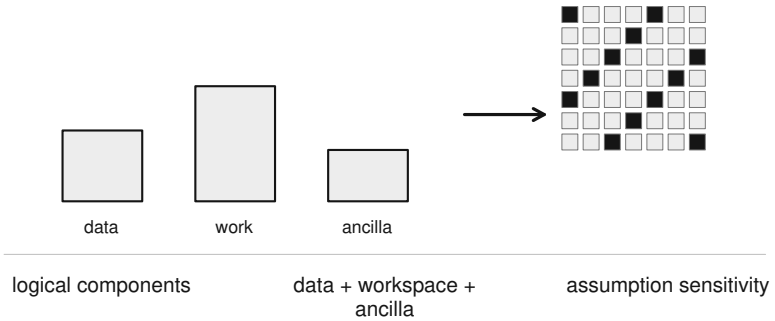
secp256k1 quantum ECDLP resource mode



Finding: Models the relevant discrete-log task separately from integer factoring; no executed target-scale quantum circuit is established.

M222 / SCHEMATIC Audited/corrected

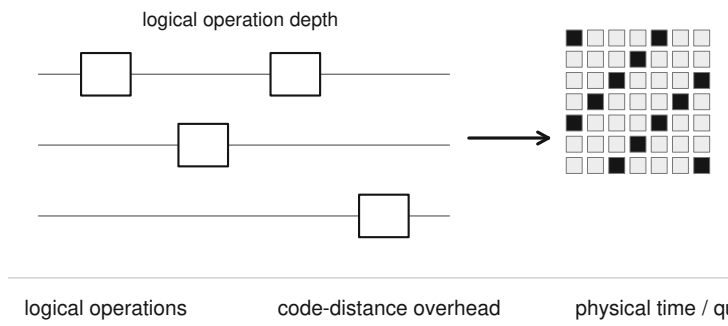
520-logical-qubit resource sensitivity model



Finding: Standalone audit excludes required reversible workspace and full architecture; headline counts are not a complete Bitcoin-breaking machine.

M223 / SCHEMATIC Implemented/model

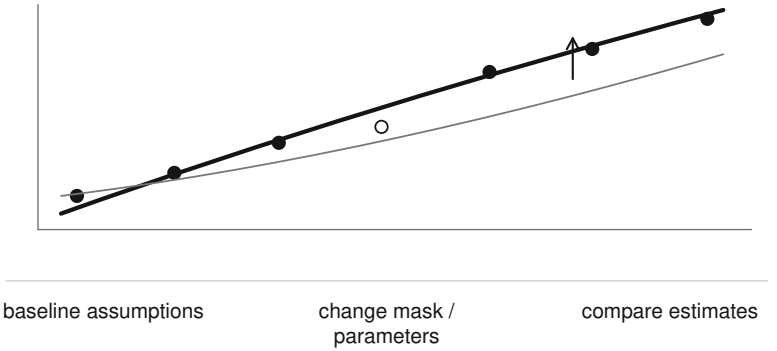
Surface-code/platform timing estimates



Finding: Compare physical errors, distances and cycle times for several platforms; estimates depend on explicit architecture assumptions.

M224 / SCHEMATIC Implemented/model

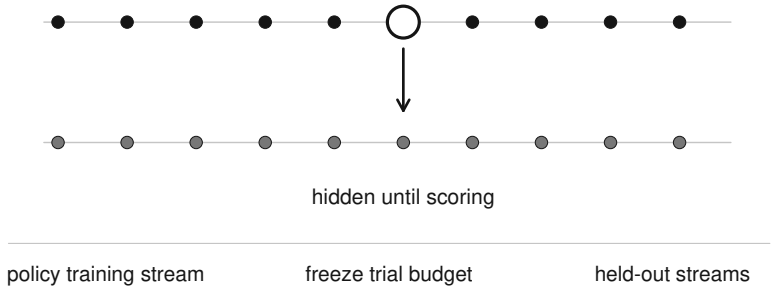
Wolfram/mask sensitivity extensions



Finding: Standalone modules explore model sensitivity and follow-up calculations, not recovered puzzle keys.

M225 / SCHEMATIC Executed record

Train/holdout adaptive-trial study



Finding: Latest factoring report includes additional controlled trial streams; conclusions remain limited to classical simulation.